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Magnetic Resonance Imaging (MRI) and Computed Tomography (CT) of the Spine

[Clinical Policy Bulletins](#) | [Medical Clinical Policy Bulletins](#)**Number: 0236**

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Policy History

[Last Review](#) [↗](#) 04/09/2026

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Next Review: 02/25/2027

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Policy

Scope of Policy

This Clinical Policy Bulletin addresses magnetic resonance imaging (MRI) and computed tomography (CT) of the spine.

Note: Precertification may be required for select procedures and interventions. Use the [CPT code search tool \(https://www.aetna.com/health-care-professionals/precertification/precertification-lists.html\)](https://www.aetna.com/health-care-professionals/precertification/precertification-lists.html) to see if precertification is required.

I. Medical Necessity

Aetna considers magnetic resonance imaging (MRI) and computed tomography (CT) of the spine medically necessary when *any* of the following criteria is met:

- Clinical evidence of spinal stenosis; *or*
- Clinical suspicion of a spinal cord or cauda equina compression syndrome; *or*
- Congenital anomalies or deformities of the spine; *or*
- Diagnosis and evaluation of lumbar epidural lipomatosis; *or*
- Evaluation of recurrent symptoms after spinal surgery; *or*
- Evaluation prior to epidural injection to rule out tumor or infection and to delineate the optimal anatomical location for performing the injection; *or*
- Follow-up of evaluation for spinal malignancy or spinal infection; *or*
- Known or suspected myelopathy (e.g., multiple sclerosis) for initial diagnosis when MRI of the brain is negative or symptoms mimic those of other spinal or brainstem lesions; *or*
- Known or suspected primary spinal cord tumors (malignant or non-malignant); *or*

Additional Information

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- Persistent back or neck pain with radiculopathy as evidenced by pain plus objective findings of motor or reflex changes in the specific nerve root distribution, and no improvement after 6 weeks of conservative therapy^{*}; *or*
- Primary spinal bone tumors or suspected vertebral, paraspinal, or intraspinal metastases; *or*
- Progressively severe symptoms despite conservative management; *or*
- Rapidly progressing neurological deficit, or major motor weakness; *or*
- Severe back pain (e.g., requiring hospitalization); *or*
- Spondylolisthesis and degenerative disease of the spine that has not responded to 4 weeks of conservative therapy^{*}; *or*
- Suspected infectious process (e.g., osteomyelitis epidural abscess of the spine or soft tissue); *or*
- Suspected spinal cord injury secondary to trauma; *or*
- Suspected spinal fracture and/or dislocation secondary to trauma (if plain films are not conclusive); *or*
- Suspected transverse myelitis.

* Conservative therapy = moderate activity, analgesics, non-steroidal anti-inflammatory drugs, muscle relaxants.

II. Experimental, Investigational, or Unproven

- Aetna considers BoneMRI (MRI-based synthetic CT) experimental, investigational, or unproven for pre-operative assessments and surgical planning of the spine and the pelvis; and for diagnosing or monitoring of tumors (primary or metastatic).
- Aetna considers MRI and CT of the spine experimental, investigational, or unproven for all other indications because their clinical value for indications other than the ones listed above has not been established. Clinical guidelines, including those from the Agency for Healthcare Policy and Research, have consistently recommended against routine imaging studies for acute low back pain.
- Aetna considers the use of MRI for further evaluation of unstable injury in neurologically intact individuals with blunt trauma after a negative cervical spine CT result not medically necessary.
- Aetna considers dynamic-kinetic MRI experimental, investigational, or unproven for evaluation of the cervical spine because its effectiveness has not been established.
- Aetna considers the use of routine MRI after a normal CT of the cervical spine in obtunded or comatose individuals experimental, investigational, or unproven because the clinical value of this approach has not been established.
- Aetna considers dual-energy CT experimental, investigational, or unproven for the evaluation of bone marrow edema and fracture lines in acute vertebral fractures because the clinical value of this approach has not been established.

III. Related Policies

- [CPB 0093 - Open Air, Low Field Strength, and Positional Magnetic Resonance Imaging \(MRI\) \(./1_99/0093.html\)](#)
- [CPB 0105 - Magnetic Resonance Imaging \(MRI\) of the Breast \(./100_199/0105.html\)](#)
- [CPB 0171 - Magnetic Resonance Imaging \(MRI\) of the Extremities \(./100_199/0171.html\)](#)
- [CPB 0202 - Magnetic Resonance Spectroscopy \(MRS\) \(0202.html\)](#)
- [CPB 0228 - Cardiac Computed Tomography \(CT\), Coronary CT Angiography, Calcium Scoring and CT Fractional Flow Reserve \(./200_299/0228.html\)](#)
- [CPB 0520 - Magnetic Resonance Imaging of the Cardiovascular System - Cardiac MRI \(./500_599/0520.html\)](#)

CPT Codes / HCPCS Codes / ICD-10 Codes

Code	Code Description
CPT codes covered if selection criteria are met:	
72125	Computed tomography, cervical spine; without contrast material
72126	with contrast material
72127	without contrast material, followed by contrast material(s) and further sections
72128	Computed tomography, thoracic spine; without contrast material
72129	with contrast material
72130	without contrast material, followed by contrast material(s) and further sections
72131	Computed tomography, lumbar spine; without contrast material
72132	with contrast material
72133	without contrast material, followed by contrast material(s) and further sections
72141	Magnetic resonance (e.g., proton) imaging, spinal canal and contents, cervical; without contrast material
72142	with contrast material(s)
72146	Magnetic resonance (e.g., proton) imaging, spinal canal and contents, thoracic; without contrast material
72147	with contrast material(s)
72148	Magnetic resonance (e.g., proton) imaging, spinal canal and contents, lumbar; without contrast material
72149	with contrast material(s)
72156	Magnetic resonance (e.g., proton) imaging, spinal canal and contents, without contrast material, followed by contrast material(s) and further sequences; cervical
72157	thoracic
72158	lumbar
Other CPT codes related to the CPB:	
76390	Magnetic resonance spectroscopy
HCPCS codes covered if selection criteria are met:	
A9575	Injection, gadoterate meglumine, 0.1 ml
A9576	Injection, gadoteridol, (ProHance multipack), per ml
A9577	Injection, gadobenate dimeglumine (MultiHance), per ml
A9578	Injection, gadobenate dimeglumine (MultiHance multipack), per ml
A9579	Injection, gadolinium based magnetic resonance contrast agent, not otherwise specified, per ml
Q9953	Injection, iron-based magnetic resonance contrast agent, per ml
Q9954	Oral magnetic resonance contrast agent, per 100 ml
ICD-10 codes covered if selection criteria are met:	
A18.01, A18.03	Tuberculosis of spine and other bones
C41.2	Malignant neoplasm of vertebral column
C41.4	Malignant neoplasm of pelvic bones, sacrum, and coccyx
C70.1	Malignant neoplasm of spinal meninges
C72.0 - C72.1	Malignant neoplasm of spinal cord
C79.51 - C79.52	Secondary malignant neoplasm of bone and bone marrow
D16.6	Benign neoplasm of vertebral column, excluding sacrum and coccyx

Code	Code Description
D16.8	Benign neoplasm of pelvic bones, sacrum and coccyx
D17.79	Benign lipomatous neoplasm of other sites [for the diagnosis and evaluation of lumbar epidural lipomatosis]
D32.1	Benign neoplasm of spinal meninges
D33.4	Benign neoplasm of spinal cord
D42.0 - D42.9	Neoplasm of uncertain behavior of meninges
D43.0 - D43.2, D43.4	Neoplasm of uncertain behavior of brain and spinal cord
D48.0	Neoplasm of uncertain behavior of bone and articular cartilage
D48.110 - D48.2	Neoplasm of uncertain behavior of connective and other soft tissue
E88.2	Lipomatosis, not elsewhere classified [for the diagnosis and evaluation of lumbar epidural lipomatosis]
G00.0 - G03.9	Meningitis
G04.00 - G04.91	Encephalitis, myelitis, and encephalomyelitis
G06.1	Intraspinal abscess and granuloma
G11.0 - G12.9	Spinocerebellar disease, anterior horn cell disease, and other diseases of spinal cord
G35.A - G35.D	Multiple sclerosis
G56.00 - G56.93 G58.0 - G58.9	Mononeuritis of upper limb and mononeuritis multiplex
G57.00 - G57.90	Mononeuritis of lower limb and unspecified site
G83.4	Cauda equina syndrome
M40.00 - M41.9	Kyphosis and lordosis
M46.20 - M46.39, M86.08, M86.18 M86.28, M86.38, M86.48, M86.58 M86.68, M86.8x8, M86.9, M89.68 M90.88	Osteomyelitis, periostitis, and other infections involving bone, other specified sites
M48.00 - M48.08	Spinal stenosis
M50.00 - M50.03 M51.04 - M51.06	Intervertebral disc disorder with myelopathy
M50.10 - M50.13	Cervical disc disorder with radiculopathy
M50.20 - M50.23, M50.90 - M50.93 M51.24 - M51.27	Displacement of intervertebral disc
M51.14 - M54.17, M54.14 - M54.17	Thoracic or lumbosacral neuritis or radiculopathy, unspecified
M54.10 - M54.18, M79.2	Neuralgia, neuritis, and radiculitis, unspecified
M54.30 - M54.32	Sciatica
M54.9	Dorsalgia, unspecified
Q04.9, Q06.9, Q07.9	Congenital malformations of brain, spinal cord, and nervous system, unspecified
Q05.0 - Q05.9	Spina bifida
Q06.0 - Q06.9	Other congenital malformations of spinal cord
Q07.00 - Q07.9	Other congenital malformations of nervous system
Q27.9	Congenital malformation of peripheral vascular system, unspecified

Code	Code Description
Q76.0 - Q76.49	Congenital malformations of spine
S12.000A - S12.9xxS S22.000A - S22.089S S32.000A - S32.2xxS	Fracture of vertebral column
S13.100A - S13.29xA S23.100A - S23.171S S33.100A - S33.39xS	Dislocation of vertebra
S14.101A - S14.159S S24.101A - S24.159S S34.101A - S34.139S	Spinal cord injury
S14.2xxA - S14.3xxS, S24.2xxA - S24.2xxA S34.21xA - S34.22xS, S34.4xxA - S34.4xxS	Injury to nerve roots and spinal plexus
ICD-10 codes not covered for indications listed in the CPB:	
R40.20 - R40.2444	Coma [not covered for use of routine MRI after a normal CT of the cervical spine]
BoneMRI (MRI-based synthetic CT):	
CPT codes not covered for indications listed in the CPB:	
G0566	3D radiodensity-value bone imaging, algorithm derived, from previous magnetic resonance examination of the same anatomy
ICD-10 codes not covered for indications listed in the CPB:	
C00.0-C14.8	Malignant neoplasm of lip, oral cavity, and pharynx
C15.3-C26.9	Malignant neoplasm of digestive organs
C30.0-C39.9	Malignant neoplasm of respiratory and intrathoracic organs
C40.00-C41.9	Malignant neoplasm of bone and articular cartilage
C43.10-C4A.122	Melanoma and other malignant neoplasms of skin
C45.0-C49.A9	Malignant neoplasm of mesothelial and soft tissue
C50.011-C50.919	Malignant neoplasm of breast
C51.0-C58	Malignant neoplasm of female genital organs
C60.0 -C63.12	Malignant neoplasm of male genital organs
C64.1 -C68.9	Malignant neoplasm of urinary tract
C69.00 -C72.59	Malignant neoplasms of eye, brain and other parts of central nervous system
C73-C75.9	Malignant neoplasm of thyroid and other endocrine glands
C76.0-C80.2	Malignant neoplasms of ill-defined, other secondary and unspecified sites
C7A.00 -C7A.098	Malignant neuroendocrine tumors
C7B.00 -C7B.09	Secondary neuroendocrine tumors
C81.00-C96.29	Malignant neoplasms of lymphoid, hematopoietic and related tissue
Z01.818	Encounter for other preprocedural examination
Z01.89	Encounter for other specified special examinations
Z08	Encounter for follow-up examination after completed treatment for malignant neoplasm
Z12.0-Z12.9	Encounter for screening for malignant neoplasms
Z15.01-Z15.09	Genetic susceptibility to malignant neoplasm
Z80.0-Z80.9	Family history of primary malignant neoplasm
Z85.00-Z85.9	Personal history of malignant neoplasm

Code	Code Description
<i>Dual-energy CT for the evaluation of bone marrow edema and fracture lines in acute vertebral fractures:</i>	
CPT codes not covered for indications listed in the CPB:	
<i>Dual-energy CT for the evaluation of bone marrow edema and fracture lines in acute vertebral fractures – no specific code</i>	
ICD-10 codes not covered for indications listed in the CPB:	
R60.0	Localized edema
S12.000A - S12.9XXS	Fracture of cervical vertebra and other parts of the neck
S22.000A - S22.089S	Fracture of thoracic vertebra
S32.000A - S32.059S	Fracture of lumbar vertebra
S32.10XA - S32.19XS	Fracture of sacrum
S32.2XXA - S32.2XXS	Fracture of coccyx

Background

Due to its complexity, the spine is likely the most challenging part of the skeletal system to assess through radiological methods. Advances in computed tomography (CT) scanners and the introduction of magnetic resonance imaging (MRI) have significantly altered the approach to spinal diagnostic imaging. In the past, invasive techniques were necessary to gather information that can now be obtained through non-invasive technologies.

MRI of the spine utilizes a strong magnetic field and radiofrequency pulses—rather than ionizing radiation—to create high-contrast, multiplanar images that are particularly effective for evaluating intervertebral discs, the spinal cord, nerve roots, the epidural space, and ligaments. It is commonly employed when there is a suspicion of neurologic injury or ligamentous/soft-tissue pathology, such as in certain trauma cases where cord or ligament injuries are a concern. In contrast, CT of the spine is an X-ray-based, cross-sectional imaging technique that uses ionizing radiation to quickly produce high-resolution images of bony structures. In cases of acute blunt spinal trauma, multidetector CT is typically the first-line imaging modality due to its speed, availability, and accuracy in detecting and classifying fractures and assessing alignment, while MRI is reserved for specific inquiries, such as suspected neurologic or ligamentous injuries that are not fully characterized by CT (ACR, 2012; Expert Panel on Neurological Imaging, 2025).

Bulging intervertebral discs are present in more than half of all asymptomatic adults. Therefore, it is essential to conduct MRI or CT at the appropriate time and to interpret the results in light of clinical findings to ensure an accurate diagnosis and to prevent unnecessary treatment of conditions that may not be responsible for a patient's symptoms.

According to accepted guidelines, MRI is the preferred method of imaging for each of the medically necessary indications listed in the Policy section, with the exception of (i) suspected spinal fracture or dislocation due to trauma, where CT scan is the preferred method of imaging if plain films are inconclusive, and (ii) evaluation of a patient with signs or symptoms of spinal stenosis, where MRI or CT are equally appropriate. For evaluation of recurrent symptoms after spinal surgery, MRI with and without gadolinium enhancement, is the preferred method of imaging.

Magnetic resonance imaging or CT evaluation of chronic mechanical low back pain (LBP) without radiculopathy or neurologic deficit, trauma, or clinical suspicion of systemic disorder (e.g., infectious process, metastatic disease) is not necessary unless back pain is severe (e.g., requiring

hospitalization) or where symptoms are progressing despite conservative management (ICSI, 2002).

The American College of Physicians (2012) has recommended against obtaining imaging studies in patients with non-specific low back pain. In patients with back pain that cannot be attributed to a specific disease or spinal abnormality following a history and physical examination (e.g., non-specific low back pain), imaging with plain radiography, computed tomography (CT) scans, or magnetic resonance imaging (MRI) does not improve patient outcomes. The American Academy of Family Physicians (2012) recommends against imaging for low back pain within the first six weeks unless red flags are present. Red flags include, but are not limited to, severe or progressive neurological deficits or when serious underlying conditions such as osteomyelitis are suspected. Imaging of the lower spine before six weeks does not improve outcomes but does increase costs. Low back pain is the fifth most common reason for all physician visits. The North American Spine Society (2013) has issued similar recommendations.

Cho et al. (2009) reported the results of a systematic review and meta-analysis of imaging strategies for low back pain (LBP) without indications of serious underlying conditions. Inclusion criteria were randomized controlled trials that compared immediate, routine lumbar imaging (or routine provision of imaging findings) versus usual clinical care without immediate lumbar imaging (or not routinely providing results of imaging) for LBP without indications of serious underlying conditions. Primary outcomes were improvement in pain or function. Secondary outcomes included improvement in mental health, quality of life, patient satisfaction, and overall improvement. Outcomes were categorized as short-term (less than or equal to 3 months), long-term (greater than 6 months to less than or equal to 1 year), or extended (greater than 1 year). A total of six trials met the inclusion criteria: four assessed lumbar radiography, and two assessed MRI or CT. Duration of follow-up ranged from 3 weeks to 2 years. One trial excluded patients with sciatica or other symptoms of radiculopathy, and one did not report the proportion of patients with such symptoms. In the other four trials, the proportion of patients with sciatica or radiculopathy ranged from 24% to 44%. Three trials compared immediate lumbar radiography with usual clinical care without immediate lumbar radiography, and one compared immediate lumbar radiography with a brief education intervention plus lumbar radiography if no improvement was seen by 3 weeks. Patients (n = 1,804) enrolled in these trials had mainly acute or subacute (less than 12 weeks) LBP, and all trials were conducted in primary care or urgent care settings. Two studies assessed advanced imaging modalities. One study compared immediate MRI or CT with usual clinical care without advanced imaging in patients with mainly chronic LBP (82% had LBP for greater than 3 months) referred to a surgeon, whereas in the other study, all patients with LBP for less than 3 weeks underwent MRI, with randomization to routine notification of results within 48 hours versus notification of results only if clinically indicated. Patients were recruited from various settings (primary care, spine clinic, or emergency room). In both trials, the proportion of patients who underwent lumbar radiography before enrollment was not reported. The most frequent methodological shortcoming was the lack of (or unclear use of) blinded outcome assessment (5 of 6 trials), followed by inadequate description of the randomization method (4 of 6 trials). All trials excluded patients with features suggestive of a serious underlying condition, but exclusion criteria varied, and trials did not indicate the number of patients excluded because of such factors. The authors found no significant difference between routine, immediate lumbar imaging and usual clinical care without immediate imaging for improvement in pain or function at short-term or long-term follow-up. In the trial that reported extended (2-year) follow-up data, immediate MRI or CT was not better than usual clinical care without immediate imaging on either the EuroQol-5D (mean difference 0.02, 95% confidence interval: -0.02 to 0.07, 0 to 1 scale) or the SF-36 mental health score (-1.50, -4.09 to 1.09, 0 to 100 scale) in unadjusted analyses. The authors concluded that lumbar imaging for LBP without indications of serious underlying conditions does not improve clinical outcomes and that clinicians should refrain from routine, immediate lumbar imaging in patients with acute or subacute LBP and without features suggesting a serious underlying condition.

In a meta-analysis, Schoenfeld et al. (2010) examined whether adding an MRI would provide useful information that alters treatment when a CT scan reveals no evidence of injury in obtunded blunt trauma patients. Published studies from 2000 to 2008 involving patients undergoing MRI for further

cervical spine evaluation after a "negative" CT scan were identified via a literature search of online databases. Data from eligible studies were pooled, and original scale meta-analyses were performed to calculate overall sensitivity, specificity, positive and negative predictive values, likelihood ratios, and relative risk. The Q-statistic p-value was used to evaluate heterogeneity. A total of 11 studies met the inclusion criteria, yielding data on 1,550 patients with a negative CT scan after blunt trauma subsequently evaluated with an MRI. The MRI detected abnormalities in 182 patients (12%). Ninety traumatic injuries were identified, including ligamentous injuries (86/182) and fractures and dislocations (4/182). In 96 cases (6% of the cohort), the MRI identified an injury that altered management. Eighty-four patients (5%) required continued collar immobilization, and 12 (1%) required surgical stabilization. The Q-statistic p-value for heterogeneity was 0.99, indicating the absence of heterogeneity among the individual study populations. The authors concluded that reliance on CT imaging alone to "clear the cervical spine" after blunt trauma can lead to missed injuries. The findings of this study supported the addition of MRI in evaluating patients who are obtunded or unexaminable, despite a negative CT scan.

Callaghan et al. (2012) examined diagnostic practice patterns as an early step in identifying opportunities to improve the efficiency of care for patients with peripheral neuropathy. The 1996 to 2007 Health and Retirement Study Medicare claims-linked database was used to identify individuals with an incident diagnosis of peripheral neuropathy using International Classification of Diseases, Ninth Revision, codes, requiring no previous neuropathy diagnosis during the preceding 30 months. Focusing on 15 relevant tests, these investigators examined the number and patterns of tests and specific test utilization 6 months before and after the incident neuropathy diagnosis. Medicare expenditures were assessed during the baseline, diagnostic, and follow-up periods. Of the 12,673 patients, 1,031 (8.1%) received a new International Classification of Diseases, Ninth Revision, diagnosis of neuropathy and met the study inclusion criteria. Of the 15 tests considered, a median of 4 (interquartile range, 2 to 5) tests were performed, with more than 400 patterns of testing. Magnetic resonance imaging of the brain or spine was ordered in 23.2% of patients, whereas a glucose tolerance test was rarely obtained (1.0%). Mean Medicare expenditures were significantly higher in the diagnostic period than in the baseline period (\$14,362 versus \$8,067, $p < 0.001$). The authors concluded that patients diagnosed with peripheral neuropathy typically undergo many tests, but testing patterns are highly variable. Almost 25% of patients receiving neuropathy diagnoses undergo high-cost, low-yield MRI, whereas few receive low-cost, high-yield glucose tolerance tests. Expenditures increase substantially in the diagnostic period. The authors stated that more research is needed to define effective and efficient strategies for the diagnostic evaluation of peripheral neuropathy.

Additionally, an UpToDate review on "Overview of polyneuropathy" (Rutkove, 2012) does not mention the use of MRI or CT in the diagnostic evaluation of individuals with polyneuropathy.

The Institute for Clinical Systems Improvement clinical practice guideline on "Adult acute and subacute low back pain" (ICSI, 2012) stated that imaging (CT, MRI, or X-ray) is not recommended for non-specific low back pain [strong recommendation, moderate quality evidence].

El Barzouhi et al. (2013) noted that MRI is frequently performed during follow-up in patients with known lumbar disk herniation and persistent symptoms of sciatica. The association between findings on MRI and clinical outcomes is controversial. These investigators studied 283 patients in a randomized trial comparing surgery and prolonged conservative care for sciatica and lumbar disk herniation. Patients underwent MRI at baseline and after 1 year. These researchers used a 4-point scale to assess disk herniation on MRI, ranging from 1 for "definitely present" to 4 for "definitely absent." A favorable clinical outcome was defined as complete or nearly complete disappearance of symptoms at 1 year. These investigators compared the proportions of patients with a favorable outcome among those with a definite absence of disk herniation and those with a definite, probable, or possible presence of disk herniation at 1 year. The area under the receiver-operating-characteristic (ROC) curve was used to assess the prognostic accuracy of the 4-point scores regarding a favorable or unfavorable outcome, with 1 indicating "perfect discriminatory value" and 0.5 or less indicating "no discriminatory value." At 1 year, 84% of the patients reported having a favorable outcome. Disk herniation was visible in 35% of those with a favorable outcome and in 33% with an unfavorable outcome ($p = 0.70$). A favorable

outcome was reported in 85% of patients with disk herniation and 83% without disk herniation ($p = 0.70$). Assessment of disk herniation by means of MRI did not distinguish between patients with a favorable outcome and those with an unfavorable outcome (area under ROC curve, 0.48). The authors concluded that MRI performed at 1-year follow-up in patients who had been treated for sciatica and lumbar disk herniation did not distinguish between those with a favorable outcome and those with an unfavorable outcome. Moreover, they stated that further research is needed to evaluate the value of MRI in clinical decision-making for patients with persistent or recurrent sciatica.

Steffens et al. (2014) systematically reviewed whether MRI findings of the lumbar spine predict future LBP in different samples with and without LBP. MEDLINE, CINAHL, and EMBASE databases were searched. Included were prospective cohort studies investigating the relationship between baseline MRI abnormalities of the lumbar spine and clinically important LBP outcomes at follow-up. These researchers excluded cohorts with specific diseases as the cause of their LBP. Associations between MRI findings and LBP pain outcomes were extracted from eligible studies. A total of 12 studies met the inclusion criteria; 6 studies presented data on participants with current LBP; 1 included a sample with no current LBP, 3 included a sample with no history of LBP, and 2 included mixed samples. Due to small sample size, poor overall quality, and the heterogeneity between studies in terms of participants, MRI findings, and clinical outcomes investigated, it was not possible to pool findings. No consistent associations between MRI findings and outcomes were identified. Single studies reported significant associations for Modic changes type 1 with pain, disc degeneration with disability in samples with current LBP, and disc herniation with pain in a mixed sample. The authors concluded that the limited number, heterogeneity, and overall quality of the studies do not permit definite conclusions on the association of MRI findings of the lumbar spine with future LBP.

Weber et al. (2015) evaluated the incremental diagnostic value of spine MRI evaluated separately from and combined with sacroiliac joint (SIJ) MRI in non-radiographic axial spondyloarthritis (nr-axSpA) compared with SIJ MRI alone. The study sample comprised two independent cohorts (A/B) of 130 consecutive patients aged less than or equal to 50 years with back pain, newly referred to two university clinics, and 20 healthy controls. Patients were classified according to clinical examination and pelvic radiographs as having nr-axSpA ($n = 50$), ankylosing spondylitis ($n = 33$), or non-specific back pain ($n = 47$). Four readers assessed SIJ and spine MRI separately 6 months apart, and 1 to 12 months later, both scans simultaneously using standardized modules. Readers recorded the presence/absence of spondyloarthritis (SpA) and their level of confidence in this conclusion on a 0 to 10 scale (0 = definitely not; 10 = definite). These researchers analyzed differences between SIJ MRI versus spine MRI alone, and SIJ MRI alone versus combined MRI, descriptively by the number/percentage of subjects according to the mean of four readers. In cohorts A/B, 15.8%/24.2% of patients with nr-axSpA having a negative SIJ MRI were reclassified as being positive for SpA by global evaluation of combined scans. However, 26.8%/11.4% of non-specific back pain controls and 17.5% of healthy volunteers with a negative SIJ MRI were falsely reclassified as having SpA by combined MRI. Low confidence in a diagnosis of SpA by SIJ MRI increased to high confidence by combined MRI in 6.6%/7.3% of patients with nr-axSpA. The authors concluded that combined spine and SIJ MRI added little incremental value compared with SIJ MRI alone for diagnosing patients with nr-axSpA and enhancing confidence in this diagnosis.

On behalf of the Tufts Medical Center Evidence-based Practice Center, Dahabreh and colleagues (2011) performed a systematic review of emerging MRI technologies for musculoskeletal imaging under loading stress for the Agency for Healthcare Research and Quality (AHRQ). The review included 57 studies about MRI under physiologic loading stress performed in an upright or sitting position or under axial loading by using a compression device. The most commonly imaged regions were the spine (33 studies) and knee (13 studies). Most studies had a cross-sectional ($n = 37$) or case-control ($n = 13$) design and reported on anatomical measurements rather than patient-relevant endpoints. Studies were generally small: the median (25th, 75th percentile) number of case patients was 26 (17, 45), and the median (25th, 75th percentile) number of control participants was 13 (12, 20 for case-control studies). Fifteen of 57 studies used at least two imaging tests and reported on diagnostic or patient-relevant outcomes but did not provide meaningful information on the relative performance of

the tests. In 10 studies that included information on adverse effects, 5% to 15% of participants reported new-onset or worsening pain and neuropathy during MRI under loading stress. The authors concluded that available evidence is insufficient to support the clinical utility of MRI under loading stress for musculoskeletal conditions.

Lord et al. (2014) reviewed the body of literature related to kinetic MRI (kMRI) of the cervical spine. A review of literature related to kMRI was performed using PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) guidelines. These researchers included 16 prospective and retrospective studies of symptomatic and asymptomatic patients who underwent kMRI of the cervical spine. The authors concluded that data suggested that kMRI is able to provide meaningful information regarding changes in the cervical spine in both normal and pathologic segments. Moreover, they stated that a prospective study comparing MRI and kMRI is needed to confirm the clinical utility of this technology.

Additionally, an UpToDate review on "Evaluation of the patient with neck pain and cervical spine disorders" (Isaac and Anderson, 2014) states that "Magnetic resonance imaging (MRI) should be the first-line imaging study performed in patients with progressive signs or symptoms of neurologic disease. MRI should also be obtained if there is a suspicion for infection or malignancy and if there are moderate to severe neck symptoms beyond six weeks, even if plain films are negative. A non-contrast MRI is sufficient in the majority of cases. The addition of gadolinium contrast intravenously allows better diagnosis of infection, tumor, or post-surgical epidural fibrosis and can be ordered subsequently if the non-contrast study is inconclusive." It does not mention the use of dynamic-kinetic MRI.

In a systematic review, Suri and colleagues (2015) examined whether lumbar muscle characteristics on MRI or CT can inform clinicians about the course of future LBP, functional limitations, or physical performance in adults with or without LBP. These investigators searched PubMed, Embase, and CINAHL through October 2014 for articles published in English in which authors assessed lumbar muscle characteristics on conventional MRI/CT as predictors of future LBP, functional limitations, or physical performance in adults. Studies with only post-surgical subjects were excluded. The search identified 3,554 articles, of which six observational cohort studies were included in the final review. These researchers used the Newcastle Ottawa Scale to evaluate potential bias. Data were extracted on study design, study population, sample size, participant characteristics, details of MRI/CT assessments, interventions, study outcomes, analysis methods, and study results. Because of heterogeneity between studies, these researchers conducted a qualitative evidence synthesis. Among high-quality studies, there was limited evidence that, for individuals with or without LBP, greater MRI-detected multifidus cross-sectional area at L5 to S1 predicted greater LBP intensity at 1-year follow-up, lesser erector spinae fat infiltration (FI) at L5 to S1 predicted greater LBP intensity at 15-year follow-up, and greater erector spinae side-to-side FI asymmetry at L3 to L4 predicted lower LBP frequency at 15-year follow-up; however, there was also limited evidence that all other MRI-detected paraspinal muscle characteristics examined were not predictive of LBP incidence, prevalence, frequency, or intensity at follow-up durations ranging from 1 to 15 years. There was limited evidence that greater CT-detected trunk muscle FI predicted worse physical performance in older adults at 3-year follow-up, but that trunk muscle cross-sectional area did not. The authors concluded that few lumbar muscle characteristics have limited evidence for an association with future LBP and physical performance outcomes, and the vast majority have limited evidence for having no association with such outcomes.

MRI / CT for Evaluation of Cervical Spinal Injury / Discontinuation of Cervical Collar Use

Veiga and Mitchell (2019) noted that a missed cervical spinal injury could have devastating consequences. Patients with a suspected cervical spinal injury are kept in rigid collars for cervical immobilization. Prolonged collar use has important clinical implications. A well-defined guideline related to the removal of cervical collars from adult obtunded blunt trauma patients has not been developed. These researchers determined if MRI offered a definitive benefit over CT with respect to patient management. They searched Ovid Online, EBSCO, NICE Evidence Journals, Medline, PubMed, BNI, CINAHL and Google Scholar as well as the grey literature. Data extraction and synthesis were performed on studies that compared the radiologic findings and clinical outcomes of CT

scan and MRI in this patient group. There is evidence that supports the safe discontinuation of cervical collar use after a negative multi-detector CT scan result alone; MRI may detect a significant number of ligamentous injuries, but such injuries are rarely of clinical significance because they rarely alter clinical management. Its use should be limited to specific circumstances.

MRI for Evaluation of Whiplash and Non-Specific Neck Pain

Owens and colleagues (2018) stated that morphometric changes to cervical musculature in whiplash-associated disorder have been reported in several studies with varying results. However, the evidence remains unclear due to the limited number of cohorts studied, with one cohort reported in multiple publications. In a systematic review with meta-analysis, these researchers evaluated the evidence for cervical muscle morphometric changes on MRI following whiplash. They searched PubMed, Medline, and the Cochrane Library without language restrictions, using combinations of the MeSH terms "muscles," "whiplash injuries," and "magnetic resonance imaging." Studies of acute and chronic whiplash were included if they compared cervical spine muscle morphometry measurements from MR images between whiplash and control groups. The search identified 380 studies, and after screening, eight studies describing five cohorts (one acute, three chronic, and one both acute and chronic) met the inclusion criteria. Participant characteristics and outcome measures were extracted using a standard format, and the quality of eligible studies was assessed using the Newcastle-Ottawa Scale. Muscle cross-sectional area (CSA) and fat infiltrate (MFI) for acute and chronic whiplash cohorts were compared using mean difference (MD) and 95% confidence intervals (CIs). Meta-analysis models were created when data from more than two eligible cohorts were available, using inverse-variance random-effects models (RevMan5 version 5.3.5). The quality assessment was uniformly good, but only two studies blinded the assessor. Analysis of the acute cohorts revealed no consensus regarding CSA, and MFI was not measured in these cohorts. In chronic cohorts, CSA was probably increased in some muscles after whiplash, but there was insufficient evidence to confirm whether MFI was also increased. Due to limited data, meta-analyses were performed only for multifidus, revealing a significant increase in chronic whiplash multifidus CSA at C5 ($Z = 3.51$, $p < 0.01$) and C6 ($Z = 2.66$, $p < 0.01$), while MFI was significantly increased at C7 only ($Z = 2.52$, $p < 0.01$); however, the heterogeneity was unacceptably high ($I^2 = 83\%$). The authors concluded that the strength of the evidence for cervical muscle morphometric changes on MRI after whiplash was inconsistent for both CSA and MFI, and they recommended that future study designs should standardize the quantification of three-dimensional muscle morphometry.

Farrell and colleagues (2019) noted that there is uncertainty regarding the clinical significance of findings on MRI in patients with whiplash-associated disorder (WAD) or non-specific neck pain (NSNP). In a systematic review and meta-analysis, these researchers compared the presence of cervical spine MRI findings in individuals with WAD or NSNP against pain-free controls. The subjects included adults with WAD ($n = 994$), NSNP ($n = 715$), or pain-free controls ($n = 2,323$). They searched Medline, Embase, CINAHL, Web of Science, SCOPUS, and Cochrane CENTRAL databases; two independent reviewers identified studies for inclusion and extracted data. The risk of bias was assessed using the Quality Assessment Tool for Observational Cohort and Cross-Sectional Studies, and the overall quality of the evidence from the meta-analysis was assessed using the GRADE approach. Meta-analysis was performed using a random-effects model to calculate odds ratios (ORs) or standard mean differences (SMDs) for binary and continuous data. A total of 31 studies were included (eight comparing acute WAD to controls, 14 comparing chronic WAD to controls, and 12 comparing chronic NSNP to controls), comprising 4,032 subjects. The rectus capitis posterior major cross-sectional area (CSA) was found to be smaller in individuals with chronic NSNP than in controls (2 studies: SMD -1.18 [95% CI: -1.65 to -0.71]). The remaining meta-analysis comparisons showed no group differences in MRI findings. The quality of evidence was mostly low due to small sample sizes and high heterogeneity. The authors concluded that, given the typically low-quality evidence, definitive conclusions could not be drawn regarding the presence of MRI findings in individuals with WAD or NSNP compared to pain-free controls.

Furthermore, an UpToDate review on "Evaluation of the adult patient with neck pain" (Isaac and Kelly, 2019) states that "MRI imaging should be performed urgently in patients suspected of having an infection, malignancy, or spinal cord compression. In the absence of red flags, imaging is not necessary in patients with mild acute or chronic neck pain that does not limit or interrupt daily activities, does not affect performance of occupation, and is easily ignored when distracted. Patients who have undergone low-velocity neck trauma (e.g., whiplash) also generally do not require imaging." The review further states, "Imaging is indicated for patients with persistent moderate to severe neck pain (e.g., lasting >6 weeks and affecting sleep or ability to perform daily activities and/or occupation) even if they lack 'red flags.' For most of these individuals (e.g., without concern for infection or malignancy, no localizing neurologic symptoms or signs, no major trauma), the preferred initial examination is cervical spine radiography. If there are concerning abnormalities noted on cervical spine radiography (e.g., endplate erosion and soft tissue swelling raising concern about discitis/osteomyelitis, bony destruction raising concern about metastases, or bony remodeling suggesting an underlying mass), cervical spine MRI without contrast should be performed. MRI is generally not indicated if radiographs are normal or show only degenerative changes."

Computed Tomography-Guided Percutaneous Spine Biopsies for Determination of a Causative Organism in Cases of Suspected Infection

Sertic and colleagues (2019) noted that in cases of suspected spondylodiscitis and vertebral osteomyelitis, CT-guided biopsies are often performed to determine the causative organism and guide antimicrobial therapy. These researchers aimed to determine the diagnostic culture yield of CT-guided biopsies performed in cases of suspected spinal infections. A literature search of PubMed and Medline up to April 2017 was conducted using the keywords "CT guided vertebral biopsy infection," "CT-guided spine biopsy infection," "CT guided spine biopsy yield," and "CT guided vertebral biopsy yield." Inclusion criteria primarily consisted of studies exclusively using CT-guided biopsies in cases of suspected infectious lesions only. After study selection, published articles were analyzed to determine diagnostic culture yield, and descriptive statistics were applied. A total of 220 search results were screened, and 11 met the inclusion criteria and were reviewed. A total of 647 biopsies of suspected infectious spinal lesions were performed, with positive cultures obtained in 241 cases. Upon excluding one paper's skewed results, the net pooled culture yield was 33%. Several cultures grew multiple organisms, leading to a total of 244 species identified. The most commonly isolated organisms included *Staphylococcus aureus* (n = 83), coagulase-negative *Staphylococcus* (n = 45), and *Mycobacteria* (n = 38). The authors concluded that the diagnostic culture yield for CT-guided biopsies in cases of suspected spinal infection was low, approximately 33%. The reasons for this low yield were likely multifactorial and have not yet been clearly defined, including the effects of pre-administration of antibiotics, biopsy technique, inadequate sample volume, suboptimal specimen transfer methods, and culture techniques. These researchers stated that further study of these individual variables is needed, with a clearly defined and universally applied standard reference method. Considering that the administration of antibiotics is often delayed in an attempt to first determine a causative organism, the question of clinical utility is raised, especially given the potential consequences of doing so. Advances in technology and hospital policy regarding specimen acquisition, tissue transfer, and handling are needed to ensure that the benefits of CT-guided biopsies outweigh the risks. Collaboration between interventional radiologists and pathologists is essential to optimize these techniques and ensure optimal results.

The authors stated that the main drawback of this review was the heterogeneity of metrics in the 11 studies. Unfortunately, it was difficult to evaluate whether any particular variable, such as biopsy method or specimen transfer and processing method, had a statistically significant effect on diagnostic culture yield across multiple papers. Another major drawback was that nearly all included studies were retrospective reviews. Lastly, as there exists an imperfect gold standard for the diagnosis of vertebral osteomyelitis, there was considerable heterogeneity in how each study defined their reference method. Some studies employed a composite reference that included microbiological diagnosis or histopathology, radiological appearance, and clinical response to antimicrobial therapy, whereas others

utilized positive microbiology or histopathology alone or heavily relied on clinician judgment. This not only made comparison between studies difficult but also complicated any measure of clinical sensitivity, specificity, or accuracy.

MRI / CT for the Diagnosis and Evaluation of Lumbar Epidural Lipomatosis

Ishikawa and associates (2006) stated that idiopathic symptomatic spinal epidural lipomatosis (SEL) is a rare condition, with few reports discussing diagnostic imaging criteria. These researchers examined factors related to clinical symptoms, correlations between clinical features and the presence of spinal epidural fat, and reviewed the literature concerning idiopathic SEL. Morphological gradings of epidural fat were assessed in seven patients with idiopathic SEL using MRI. Additionally, body mass index (BMI), the number of involved vertebral levels, grade, and pre-operative Japanese Orthopedic Association (JOA) scores were analyzed. Surgery resulted in symptomatic relief, with a mean JOA score recovery rate of 67.4%. Grading of epidural fat tended to show a slight negative correlation with pre-operative JOA scores, while a strong significant positive correlation was found between the number of involved vertebral levels and BMI. The authors concluded that the number of involved vertebral levels and obesity were strongly correlated, whereas the severity of dural compression was not always significantly associated with neurological complications. They indicated that epidural fat in the lumbar spine contributed to neurological deficits and suggested that weight-reduction therapy could decrease the number of vertebral levels involved. MRI-based grading was deemed helpful for the diagnosis and evaluation of idiopathic lumbar SEL, and symptoms and neurological findings were important for determining the surgical approach.

Gerardy and colleagues (2021) noted that SEL is characterized by excessive accumulation of normal fat in the epidural space. They presented findings from a retrospective study of 20 patients (17 men and 3 women) with an average age of 64 years, all suffering from radiculopathy and/or neurogenic claudication. Lipomatosis was idiopathic in six patients and secondary in 14 patients. MRI grading revealed grade-II lipomatosis in 30% of cases and grade-III in 70% of cases. All subjects improved following decompressive surgery via laminectomy and resection of epidural fat. The authors concluded that surgical decompression is a safe and effective procedure for patients with symptomatic lumbar epidural lipomatosis, particularly in cases of failed conservative treatment or neurological deficits.

Walker and co-workers (2021) stated that SEL is caused by an excess of adipose tissue accumulation localized to the thoracic and lumbar regions of the spine. While the exact pathogenesis remains unclear, the etiology of SEL can be broadly classified into five commonly associated risk factors: exogenous steroid use, obesity, endogenous steroid hormonal disease, spine surgery, and idiopathic disease. Progression of SEL may lead to neurological deficits, myelopathy, radiculopathy, neurogenic claudication, loss of sensation, difficulty voiding, lower extremity weakness, and, rarely, cauda equina syndrome. The diagnosis of SEL can be challenging, as symptoms often resemble other common conditions such as vertebral and disc disease. Diagnosis requires a high degree of clinical suspicion and various diagnostic modalities, with advanced imaging, specifically T1-weighted MRI, being the most sensitive test. Grading patterns of SEL were defined based on the epidural fat (EF) to spinal column (Spi C) index, with normal being less than or equal to 40%, and grading of SEL described as grade I, II, or III, with grade III characterized by an EF/Spi C index of greater than or equal to 75%.

Spinnato et al. (2022) noted that SEL is defined as the abnormal accumulation of unencapsulated adipose tissue in the spinal epidural space. SEL can be asymptomatic or cause a wide range of symptoms, the most common being neurogenic claudication. Other neurological manifestations may include myelopathy and radicular symptoms. The lumbar region is the most frequently involved spinal level, followed by the thoracic region. Imaging plays a crucial role in disease assessment, with MRI considered the most effective and sensitive modality for diagnosing and staging SEL; however, CT scans can also diagnose SEL. The diagnosis may be incidental in mild to moderate cases or considered in cases with neurological symptoms in moderate to severe disease. Recognized risk factors for SEL include exogenous steroid use and obesity, with recent studies finding associations between SEL and hyperlipidemia and liver fat deposition. SEL can be viewed as a spinal hallmark of

metabolic syndrome. Initial treatment strategies focus on controlling risk factors (e.g., weight loss, cessation of steroid therapy), while surgical decompression may be necessary when conservative treatment fails or when acute/severe neurological symptoms develop.

Furthermore, an UpToDate review on "Polyradiculopathy: Spinal stenosis, infectious, carcinomatous, and inflammatory nerve root syndromes" (Rutkove and Tarulli, 2022) states that "Additional causes of spinal stenosis include congenital conditions, such as hereditary spinal stenosis or achondroplasia, systemic conditions such as ankylosing spondylitis or Paget disease, and excess accumulation of epidural fat (epidural lipomatosis), which can be seen in patients with Cushing disorder, exogenous glucocorticoid exposure, or obesity. Magnetic resonance imaging (MRI) has become the test of choice in the evaluation of spinal stenosis, although bony changes such as osteophyte formation are better demonstrated by computed tomography (CT)."

Upright MRI for Evaluation of the Spine

Kern et al. (2019) noted that the treatment of patients with spinal stenosis and concurrent degenerative spondylolisthesis is controversial, with two large randomized controlled trials (RCTs) reporting contradictory results. The researchers hypothesized that a substantial number of patients would show evidence of micro-instability after a sole decompression procedure. This study was a retrospective analysis of all cases of lumbar spinal stenosis treated at the Frankfurt University Clinic from 2010 through 2013. Patients with associated spondylolisthesis underwent upright magnetic resonance imaging (MRI) studies in flexion and extension to identify subtle signs of micro-instability. Clinical outcomes were assessed using the SF-36 bodily pain (BP) and physical functioning (PF) scales. A total of 21 patients were recruited for upright MRI studies, with a mean follow-up duration of 65 months (SD 16 months). Of these 21 patients, 10 (47%) showed signs of micro-instability, defined as movement greater than 4 mm on flexion/extension MRI. Comparison of mean SF-36 BP and PF scores between patients with and without micro-instability showed no statistically significant difference on either scale. The authors concluded that a substantial subset of patients developed morphological micro-instability after sole decompression procedures but did not experience any clinically significant effects from the instability.

Berry et al. (2019) stated that understanding changes in lumbar spine (LS) angles and intervertebral disc (IVD) behavior in end-range positions in healthy subjects could provide a basis for developing more specific LS models and comparing individuals with spine pathology. These researchers quantified three-dimensional (3D) LS angles and changes in IVD characteristics with end-range positions in three planes of motion using upright MRI in healthy individuals, determining which intervertebral segments contributed most in each plane of movement. A total of 13 participants (average age 24.4 years, range 18 to 51 years; 9 females; body mass index [BMI] = 22.4 ± 1.8 kg/m²) with no history of low back pain (LBP) were scanned in an upright MRI in various positions: standing, sitting flexion, sitting axial rotation (left, right), prone on elbows, prone extension, and standing lateral bending (left, right). Global and local intervertebral LS angles were measured, along with the anterior-posterior (AP) length of the IVD and the location of the nucleus pulposus. In the sagittal plane, lower LS segments contributed most to changes in position, with the nucleus pulposus migrating from a more posterior position in sitting flexion to a more anterior position in end-range extension. For lateral bending, the upper LS contributed most to end-range positions. Small degrees of intervertebral rotation (1 to 2°) were observed across all levels for axial plane positions, with no systematic changes in IVD characteristics for axial or coronal plane positions.

Papavero et al. (2020) stated that redundant nerve roots (RNRs) are a negative prognostic factor in patients with central lumbar spinal stenosis (LSS), with 40% of candidates for surgical decompression showing RNRs (RNR+) on pre-operative conventional MRI. In a retrospective observational study, these investigators examined the prevalence of RNRs in three functional postures (standing, neutral sitting, and flexed sitting) using upright MRI (upMRI). A total of 30 surgical candidates underwent upMRI, and sagittal and axial T2-weighted images of the three functional postures were evaluated. The segmental length of the lumbar spine (sLLS), lordotic angle (LA), and dural cross-sectional area (DCSA) were measured in each body position. Generalized linear mixed models were performed, with

a significance level set at 0.05. The prevalence of RNRs decreased from 80% during standing to 16.7% during flexed sitting ($p < 0.001$). The sLLS increased significantly from standing to neutral sitting in both RNR groups ($p < 0.001$), while the increase from neutral sitting to flexed sitting was only significant ($p < 0.001$) for the group without RNRs (RNR-). The LA decreased significantly for both RNR groups from standing to flexed sitting ($p < 0.001$). The DCSA increased significantly in the RNR- group ($p < 0.001$) but not in the RNR+ group ($p = 0.9$). The authors concluded that the prevalence of RNRs was body position-dependent, and increases in DCSA played a determinant role in resolving RNRs.

Shaikh et al. (2020) examined the effect of upright, seated, and supine postures on lumbar muscle morphometry at multiple spinal levels and for multiple muscles. A total of six asymptomatic volunteers were imaged using a 0.5 T upright open MRI in seven postures (standing, standing holding 8 kg, standing 45° flexion, seated 45° flexion, seated upright, seated 45° extension, and supine), with scans at L3/L4, L4/L5, and L5/S1. Muscle cross-sectional area (CSA) and muscle position relative to the vertebral body centroid (radius and angle) were measured for the multifidus/erector spinae combined and psoas major muscles. Posture significantly affected the multifidus/erector spinae CSA, with a decrease in CSA from straight postures (standing and supine) to seated and flexed postures (up to 19%). Psoas major CSA significantly varied with vertebral level, showing opposite trends due to posture at L3/L4 (increasing CSA, up to 36%) and L5/S1 (decreasing CSA, up to 40%) with sitting/flexion. For both muscle groups, radius and angle followed similar trends, with a decreasing radius (up to 5%) and increasing angle (up to 12%) in seated/flexed postures. CSA and lumbar lordosis showed some correlation (multifidus/erector spinae L4/L5 and L5/S1, $r = 0.37$ to 0.45 ; psoas major L3/L4 left, $r = -0.51$). There was generally good repeatability (average ICC (3, 1): posture = 0.81, intra = 0.89, inter = 0.82). The authors concluded that changes in multifidus/erector spinae muscle CSA likely represented muscles stretching between upright and seated/flexed postures. For the psoas major, the differential level effect suggested that changing 3D muscle morphometry with flexion was not uniform along the muscle length. The muscle and spinal level-dependent effects of posture and spinal curvature correlation, including muscle CSA and position, highlighted the importance of considering measured muscle morphometry from different postures in spine models.

In an observational study, Rustagi et al. (2020) examined differences in spine structure measures between experimental postures and standard supine posture MRIs. A total of 34 patients with low back pain (LBP) were included, and MRIs were taken in six experimental postures. The dependent measures included sagittal view anterior (ADH), middle and posterior disc heights, thecal sac width, and left/right foraminal height (FH). In the axial view, measures included disc width and left and right foraminal height, assessed at L3/L4, L4/L5, and L5/S1. Each subject served as their own control, and spine measurements in the experimental posture were compared to the same measures in the standard supine posture. A 94% inter-observer reliability was observed. In the sagittal and axial views, 55 of the 108 and 11 of the 18 measures were significantly different. In the sagittal view:

- ADH was significantly smaller in the sitting flexed posture by $2.50 \text{ mm} \pm 0.63$ compared to the supine posture;
- ADH in sitting neutral posture was significantly smaller than the standard posture by $1.97 \text{ mm} \pm 0.86$;
- sitting flexed posture showed that bilateral FH measures were significantly different;
- Bilateral FH was larger in the sitting neutral posture compared to the standard supine posture by $0.87 \text{ mm} \pm 0.17$.

The authors concluded that this research quantified the differences in spine structure measures that occurred in various experimental postures. The additional information gathered from an upright MRI may correlate with symptoms leading to an accurate diagnosis and assist in future spine research.

Bone Density Measurements on CT or MRI versus DEXA Scan

Ahmad et al. (2023) stated that novel methods of bone density assessment using CT and MRI have been increasingly reported in the spine surgery literature. Correlations between these newer measurements and traditional dual energy X-ray absorptiometry (DXA) is unclear. In a systematic

review, these investigators examined correlations between BMD from CT or MRI and DXA. Studies published between 2011 and 2021 that reported correlations between the CT- Hounsfield unit (HU) or MRI measurements to DXA t-scores or BMD of lumbar spine or hip were included in this systematic review. A total of 25 studies (15 CT, 10 MRI) met the inclusion criteria with a total number of 2,745 patients. The pooled correlation coefficient of spine CT-HU versus spine DXA, spine CT-HU versus hip DXA and spine CT-HU versus lowest t-score were 0.60, 0.50 and 0.60, respectively. Regarding spine DXA parameters, the pooled r^2 for spine CT-HU versus spine t-score was 0.684 and spine CT-HU versus spine BMD was 0.598. In addition, in patients undergoing spine surgery in 4 studies, the pooled correlation between spine CT and spine DXA was (r^2 : 0.64). In MRI studies, the pooled r^2 of spine MRI versus spine DXA and spine MRI versus hip DXA were -0.41 and -0.44, respectively. The authors concluded that CT-HU exhibited stronger correlations with DXA than MRI measurements. Lumbar CT-HU has the highest pooled correlation ($r^2 = 0.6$) with both spine DXA and lowest skeletal t-score followed by lumbar CT-HU with hip DXA ($r^2 = 0.5$), and lumbar MRI with hip ($r^2 = 0.44$) and spine ($r^2 = 0.41$) DXA. Both imaging modalities achieved only a moderate correlation with DXA. Few studies in both modalities have examined the correlation in spine surgery populations and the available data demonstrated that the correlations were worse in the degenerative spine population. A careful interruption of CT-HU and MRI measurement when evaluation of BMD as they only moderately correlated with DXA scores. At this time, it is unclear which modality is a better predictor of mechanical complications and clinical outcomes in spine surgery patients.

Osteoporotic Vertebrae Fracture Risk Measured by Quantitative Computed Tomography

In a systematic review and meta-analysis, Chen et al. (2023) compared the use of quantitative CT (QCT) and DXA in the evaluation of osteoporosis with vertebral fracture and osteoporosis without fracture. It has been reported that the volumetric BMD (vBMD) measured by QCT exhibits a stronger correlation with fracture risk than areal BMD (aBMD) measured by DXA. These investigators examined the ability of QCT and DXA to distinguish between osteoporosis with vertebral fracture and osteoporosis without fracture according to vBMD and aBMD. They carried out a primary literature search of the online databases up to July 3, 2022, in both English and Chinese publications, combining synonyms for "QCT", "DXA", and "osteoporosis". The Newcastle-Ottawa scale (NOS) was used to assess the quality of the selected articles. vBMD obtained via QCT and aBMD obtained via DXA were extracted, and were analyzed by Review Manager 5.4 and RStudio. A total of 6 studies with 610 individuals aged 45 to 90 years, of which 179 had vertebral fractures, were included in the final analysis. The weighted MD (WMD) between osteoporosis with vertebral fracture and osteoporosis without fracture for vBMD was - 27.08 (95% CI: - 31.24 to - 22.92), while for aBMD was - 0.05 (95% CI: - 0.08 to - 0.03). The authors concluded that both vBMD detected by QCT and aBMD detected by DXA could discriminate fracture status in the spine, and vBMD performed a stronger correlation with fracture risk.

Comparative Effectiveness of 4 Techniques for Identifying Vertebral Fragility Fractures

Ma et al. (2025) noted that the incidence of vertebral fragile fractures peaked among the elderly population, and identifying individuals at high risk of vertebral fractures and promptly instituting preventions are of critical importance. These investigators examined the effectiveness and values of Hounsfield unit (HU) values, vertebral bone quality (VBQ) scores, BMD measured by DXA, and QCT to discriminate between patients with and without vertebral fractures. They carried out a thorough search across PubMed, Embase, Cochrane Library, Web of Science, CNKI, and Wan Fang Database to identify potential studies that met the eligibility criteria. Studies that examined the use of HU values, VBQ scores, QCT-measured BMD, and DXA-measured BMD in discriminating vertebral fractures were qualified. The combined results showed that there were significant differences in HU values, VBQ scores, QCT-measured BMD, and DXA-measured BMD between the fracture and non-fracture groups. Moreover, the pooled sensitivity, specificity, and AUC of HU values were 0.82, 0.67, and 0.76, respectively; the pooled sensitivity, specificity, and AUC of VBQ scores were 0.70, 0.75, and 0.78; the pooled sensitivity, specificity, and AUC of QCT-measured BMD were 0.85, 0.76 and 0.88. The authors concluded that all 4 methods, namely HU values, VBQ scores, QCT-measured BMD, and DXA-measured BMD can effectively distinguish between patients with and without vertebral fragile fractures.

Among these, QCT-measured BMD exhibited a relatively high effectiveness in discriminating vertebral fractures. VBQ scores and HU values showed comparable effectiveness for discriminating vertebral fractures among elderly patients.

Imaging Techniques for Early Diagnosis and Management of Axial Spondyloarthritis

Jeon and Min (2024) presented recent finding of imaging studies employed for early diagnosis and management of patients with axial spondyloarthritis (axSpA). Using low-dose whole spine CT (CT syndesmophyte score [CTSS]) improved diagnostic accuracy for evaluating spinal structural progression than previous method (modified Stoke Ankylosing Spondylitis Spinal Score [mSASSS]) in axSpA. The novel definition of positive finding of sacroiliac joint (SIJ) and spine MRI allowed diagnosis of axSpA earlier than plain radiography. Furthermore, novel MRI protocol such as volumetric interpolated breath-hold examination improved detection rate of structural change of axial joints in axSpA. Nuclear medicine imaging showed potential for diagnosis and predicting progression of axSpA. Ultrasonography (US)-guided injection is useful for controlling local joint pain of axSpA. AxSpA is characterized by pain and inflammation of axial joints such as the SIJ and spine. Detection of active inflammatory lesions using MRI has expanded the subtypes of axSpA to include non-radiographic axSpA (nr-axSpA). Additionally, many other imaging techniques have improved, and can now detect structural and early inflammatory lesions of the axial joints. Furthermore, a method for quantitative measurement of syndesmophytes by CTSS has been developed; this method is more accurate and sensitive than the mSASSS for detecting spinal structural damage.

BoneMRI (MRI-Based Synthetic CT)

BoneMRI is an artificial intelligence (AI)-enabled software that provides radiation-free, CT-like images through MRI scans for all spine and pelvic anatomies, with sub-millimeter quantitative equivalence to CT scans, for both adults and adolescents. It was cleared by the FDA via the 510(k) process. It can be used to visualize the bone structures in MRI images with enhanced contrast with respect to the surrounding soft tissues. It is used in the pelvis region, which includes the bony anatomy of the sacrum, hip bones, and femoral heads, as well as the lumbar spine region, which includes the bony anatomy of the vertebrae from L3 to S1. BoneMRI is not indicated for diagnosing or monitoring tumors (primary or metastatic).

Jans et al. (2021) examined the diagnostic performance of MRI-based synthetic CT (sCT) in detecting structural lesions in the sacro-iliac (SI) joints compared to routine T1-weighted MRI. These researchers found that sCT outperformed MRI in detecting erosions, sclerosis, and ankylosis, with higher sensitivity and specificity. The authors concluded that sCT could significantly enhance the diagnostic accuracy for patients suspected of having sacroiliitis, making it a valuable tool in clinical practice.

Morbee et al. (2021) examined the geometric measurements of MRI-based sCT of the lumbar spine in comparison to conventional CT for surgical planning. The study showed that sCT provided equivalent measurements for various bony structures, with excellent inter- and intra-reader reliability. The authors concluded that these findings supported the use of sCT as a viable option for surgical planning, offering the advantage of reduced radiation exposure while maintaining diagnostic accuracy.

Staatjes et al. (2021) examined the potential of MRI-based sCT for surgical planning in the lumbar spine, presenting a proof-of-concept (POC) study entailing 3 cases. The study showed that sCT could generate CT-like images from MRI data, allowing for accurate visualization of both normal and pathological structures. The results indicated that sCT could potentially reduce the need for traditional CT scans, thus minimizing radiation exposure. The authors concluded that sCT could be a valuable tool for surgical planning and intra-operative navigation in lumbar spine procedures.

Florkow et al. (2022) evaluated the accuracy of sCT in the 3D evaluation of hip joint morphology compared to conventional CT. This trial included 30 male patients; these researchers found that sCT measurements were statistically equivalent to those obtained from CT, with high inter-observer and

intra-observer reliability. The authors concluded that these findings suggested that sCT could effectively examine hip morphology, providing a radiation-free alternative for clinical evaluations.

In a non-inferiority study, Van der Kolk et al. (2022) compared the image quality of deep learning (DL)-based sCT of the cervical spine to conventional CT. This trial involved 25 subjects and evaluated various bony structures. These researchers found that sCT was non-inferior to CT in terms of overall image quality and specific structural assessments. The authors concluded that these findings showed that sCT could visualize bony structures in the cervical spine, suggesting its potential as a radiation-free alternative to conventional CT.

Saveyn et al. (2022) discussed the potential of sCT images to enhance the evaluation of bony lesions on MRI, especially in the context of fracture assessment. The authors emphasized the need for further validation of synthetic CT in clinical practice, highlighting its promise in improving diagnostic accuracy for bony lesions while reducing radiation exposure associated with traditional CT imaging.

Morbee et al. (2022) examined the equivalency of MRI-based sCT to conventional CT in evaluating bony morphology of the hip joint. The study included adult patients who underwent both MRI and CT, with sCT images generated from MRI data. The results revealed strong agreement between sCT and CT measurements, indicating that sCT could serve as a reliable alternative for assessing hip morphology without the associated radiation exposure of CT. The authors concluded that these findings suggested that sCT may be beneficial for pre-operative assessments and surgical planning.

In a case-series study, Upadhyay et al. (2023) examined the use of sCT in evaluating lesions in children with rare musculoskeletal diseases. This study highlighted the ability of sCT to accurately identify musculoskeletal lesions in 2 patients, showing its potential as a non-radiative imaging modality. The authors concluded that these results suggested that sCT could be a valuable tool for monitoring musculoskeletal abnormalities in pediatric patients, especially in cases where traditional CT is limited due to radiation exposure concerns.

In a retrospective study, Morbee et al. (2023) determined the prevalence of incidental findings on SI joint MRI and the added value of MRI-based sCT in detecting these findings. This trial included 210 patients and found that incidental findings were more common than sacroiliitis, with sCT proving essential for diagnosing many lesions not visible on conventional MRI. The authors concluded that these findings underscored the importance of sCT in enhancing diagnostic accuracy and identifying clinically significant incidental findings. Moreover, these researchers stated that further quantitative research is needed to demonstrate the impact on the workflow, costs, and outcomes.

The authors stated that this study had several drawbacks. First, all subjects were imaged due to symptoms, giving these investigators no normal control group, although this resembled clinical practice. Second, only 210 subjects were included in this study, thus limiting the reporting of less frequent pathologies in the study population. Third, there was a risk of researcher bias, as the readers were aware of the objective of the study and reviewed MRI and sCT simultaneously for each case. In that way, the performance of sCT for detecting incidental findings could have been overestimated. Future studies should be designed to make diagnoses of incidental findings based on sCT and MRI separately. Fourth, in this study, no quantification or measurable data were available showing the impact of sCT on the workflow, costs, or outcomes. In addition, the MRI data were acquired with a high-performance 3-T MRI system, but it is not yet shown if similar image quality of sCT can be achieved with MRI systems of different field strengths and different technical performance; further studies are needed to compare the diagnostic performance of sCT with conventional CT.

In a cadaveric study, Davidar et al. (2023) presented a case showing the use of robot-assisted screw fixation using MRI-based sCT for radiation-free spine surgery. The study compared morphometric analyses of conventional CT and sCT scans, showing that sCT yielded comparable results in screw placement accuracy. The authors concluded that this study highlighted the potential of sCT to facilitate safe and effective surgical planning while minimizing radiation exposure.

Vereecke et al. (2023) examined the prevalence of anatomical variations of SI joints using MRI and sCT images in 215 patients suspected of sacroiliitis. The findings showed that 82.8% of the evaluated joints exhibited at least 1 anatomical variant, with the most common being the ilio-sacral complex and bipartite iliac bony plate. The study also identified a new variant, an accessory facet of the SI joint. Notably, the occurrence of these variants was associated with factors such as age, sex, and BMI. The authors concluded that these findings suggested that synthetic CT was effective in detecting subtle anatomical variations, which may represent subtypes of normal anatomy.

Shcherbakova et al. (2024) carried out a feasibility study on a short MRI protocol specifically designed for adolescent idiopathic scoliosis (AIS), which combines MRI and MRI-based sCT images to visualize both soft and bony structures in a single, radiation-free examination. This trial included 18 adult volunteers and examined the protocol's ability to visualize critical anatomical landmarks needed for detecting 3D pathoanatomical changes and surgical planning. The authors concluded that the results indicated high precision and reliability in measurements, suggesting that this innovative protocol could enhance the study of scoliosis and improve surgical navigation.

Iwasaka-Neder et al. (2024) compared MRI-based sCT to conventional CT (cCT) for evaluating hip morphology and maturity in pediatric patients. This trial included 38 hips from 19 patients and showed that sCT provided equivalent measurements to cCT, with high inter- and intra-observer reliability. The accuracy of sCT in determining physeal maturity status was found to be 98.4%. The authors concluded that these findings suggested that sCT is a reliable alternative to cCT for evaluating hip morphology in children, offering the advantage of reduced radiation exposure.

Schietecatte et al. (2024) examined the equivalency of MRI-based sCT to cCT in assessing the bony morphology of SI joints in children. The study included 10 participants and found that sCT provided good overall image quality and excellent agreement for various bony assessments. The authors concluded that by all the measures in a group of normal children, sCT was visually equivalent to CT for assessment of the bony morphology of pediatric SI joints. These researchers stated that in a clinical setting, adding sCT to the MRI protocol may have additional value as it allows for easily visually interpretable visualization of the bony structures of the SI joints in children without the use of ionizing radiation. Moreover, these investigators stated that future studies in larger groups of children, also including pathological scans of the SI joint, are needed to further examine the added value of sCT for assessment of structural lesions (erosions, ankylosis, and sclerosis) of sacroiliitis on sCT in pediatric SI joints in comparison with the classically used MRI sequences and CT as the gold standard where available.

The authors stated that this study had several drawbacks. The principal drawback was that these researchers did not image children with sacroiliitis; therefore, this trial was limited to assessing image quality for normal appearances and normal variation. Second, CT and MRI imaging did not take place on the same day; the additional MRI scan was obtained within approximately 1 month after the CT scan. Third, different kinds of CT, including PET-CT, were employed, depending on the clinical question.

Willesen et al. (2024) examined the effectiveness of MRI-based sCT in detecting new bone formation in patients with axial spondyloarthritis (axSpA) compared to low-dose CT (ldCT) and radiography. The study involved 17 patients and revealed that sCT showed high specificity and sensitivity for detecting structural changes, outperforming radiography. The authors concluded that these findings suggested that sCT could be a valuable tool for monitoring structural spine damage in axSpA patients, especially in clinical trials.

Krabbe et al. (2024) compared the detection of structural lesions in the SI joints using T1-weighted 3D spoiled gradient echo MRI and MRI-based sCT against conventional T1-weighted turbo spin echo MRI. This trial included 19 patients and found that both T1-weighted GRE and sCT improved sensitivity and reader confidence for detecting erosions, sclerosis, and ankylosis compared to T1-weighted TSE. The authors concluded that both 1-mm semi-coronal T1w-gradient echo (GRE) and 1-mm semi-coronal

sCT increased sensitivity and reader confidence for the detection of erosion, sclerosis, and ankylosis, compared with standard 4 mm T1w-turbo spin-echo (TSE). These researchers stated that these methods might be a useful addition to SI joint MRI protocols both in routine clinical care and as structural outcome measures in clinical trials.

The authors stated that this study had several drawbacks. First, the sample size was small ($n = 19$), which limited the ability to reveal significant differences between methods; however, these investigators were able to show several statistically significant findings that suggested that improved image resolution with T1w-GRE and sCT are promising methods to improve the detection of structural lesions compared with lower resolution T1w-TS. Second, little training was provided to the readers to allow them to become acquainted with the different types of images and apply the scoring methodology, and all readers had little experience with the T1w-GRE and especially sCT images, which may have limited the inter-reader agreement. But strikingly, the inter-reader agreement was significantly higher or tended to be higher for T1w-GRE and sCT (images that were new to all readers) compared with T1w-TSE for most lesion types. Third, the study was not designed to have a reference standard; CT could have been used as such. However, even low-dose CT may be considered inferior to an optimized T1-weighted MRI protocol with sufficient resolution and tissue contrast, and CT was not carried out since it was not considered necessary for the objectives of this trial. In a few joints, ankylosis was detected by sCT with poor or moderate confidence, but not by T1w-GRE. These researchers were unable to prove if this represented true ankylosis or if it was a false-positive finding on sCT, which would correspond to a higher specificity of T1w-GRE. Fourth, these investigators employed the same lesion definitions as Jans et al. (2021) to reproduce the image assessment methodology of that study, well aware that other groups, including the ASAS MRI group, have slightly different wording of the definitions, e.g., the ASAS MRI group also discriminated between erosion and "fat metaplasia in an erosion cavity" ("backfill"). This might have resulted in minor inconsistencies among the expert readers that were mostly used to applying the ASAS definitions; however, these investigators considered it unlikely to have had a major impact on the comparisons between sequences. Fifth, the study did not attempt to examine the accuracy or confidence in determining the absence of lesions, only their presence. Since no control subjects were included, readers may have had a bias toward detecting lesions with high sensitivity. The authors' MRI protocol for T1w-TSE had a voxel size of $4.0 \text{ mm} \times 0.8 \text{ mm} \times 0.7 \text{ mm}$, whereas other institutions employ a lower slice thickness of 3 mm. In-plane resolution may also differ depending on local protocol optimization. It was possible that a higher resolution T1w-TSE might have increased sensitivity and reader confidence for detecting structural lesions, at the expense of extended scan time; therefore, these results may not be generalizable to centers that use a significantly higher resolution of T1w-TSE. Furthermore, these findings may not be generalizable to newly diagnosed bio-naïve patients, who are likely to have somewhat less damage than the patients included in this trial, although these researchers expected the overall trends in lesion detection across imaging methods to be rather similar.

Michael et al. (2025) presented a case report detailing the use of MRI-generated sCT in the treatment of a 14-year-old male athlete with chronic bilateral spondylolysis. After unsuccessful non-operative management, the patient underwent a novel MRI that showed nonunion of the spondylolysis and persistent edema. Subsequently, he received minimally invasive bilateral L4 robot-navigated intra-laminar screw fixation, resulting in complete healing as confirmed by CT 3 months post-surgery. The authors concluded that this case highlighted the potential of MRI-generated sCT in pediatric spondylolysis management, providing a comprehensive assessment of bony and soft tissue pathology without the ionizing radiation associated with traditional CT.

Evaluation of Patients With Thoracolumbar Spine Trauma

The Congress of Neurological Surgeons' systematic review and evidence-based guidelines on "The evaluation and treatment of patients with thoracolumbar spine trauma: Radiological evaluation" (Qureshi et al, 2019) stated that "there was insufficient evidence that MRI can help predict clinical outcomes in patients with acute traumatic thoracic and thoracolumbar spine injuries".

MRI and Cervical Spine Clearance of Neurologically Intact Patients With Blunt Trauma

In a meta-analysis, Malhotra and co-workers (2017) quantified the rate of unstable injuries detected by MRI that were missed on CT in blunt cervical spine (CS) trauma patients and evaluated the utility of MRI in CS clearance. These researchers conducted a systematic review of worldwide evidence across five major medical databases. Studies were included if they reported the number of unstable injuries or provided sufficient details for inference. Variables assessed included severity, CT/MRI specifications, imaging timing, and outcome/follow-up. Pooled incidences of unstable injury on follow-up, weighted by the inverse of variance among all included patients, both obtunded and alert, were reported. Of 428 unique citations, 23 were deemed eligible, involving 5,286 patients and reporting 16 unstable injuries across five studies. The overall pooled incidence was 0.0029%. Among studies reporting only obtunded patients, the pooled incidence was 0.017%, while in alert patients, it was 0.011%. All reported positive findings were critically reviewed, and only 11 could be considered truly unstable. The authors concluded that there was significant heterogeneity in the literature regarding the use of imaging after a negative CT. The finding rate for unstable injury on MRI was extremely low in both obtunded and alert patients. They stated that although MRI is frequently performed, its utility and cost-effectiveness require further study. Key points of this meta-analysis included the following: there were 16 unstable injuries on follow-up MRI among 5,286 patients, the positive finding rate among obtunded patients was 0.12%, and the positive finding rate among alert patients was 0.72%. MRI exhibited a high false-positive rate, and its utility necessitated further studies. The use and role of "confirmatory" tests showed wide variations.

Wu and associates (2018) noted that the use of MRI for cervical clearance after a negative cervical CT scan result in alert patients with blunt trauma who are neurologically intact is not infrequent, despite poor evidence regarding its utility. These investigators evaluated the utility and cost-effectiveness of using MRI versus no follow-up in this patient population. A modeling-based decision analysis was performed during the lifetime of a 40-year-old individual from a societal perspective. The two strategies compared were no follow-up and MRI. A Markov model with a 3% discount rate was used, incorporating parameters from the literature. Base cases, along with probabilistic and sensitivity analyses, were performed to assess the cost-effectiveness of the strategies. The cost of MRI follow-up was \$11,477, with a health benefit of 24.03 quality-adjusted life-years (QALY); the cost of no follow-up was \$6,432, with a health benefit of 24.08 QALY. No follow-up emerged as the dominant strategy, offering a lower cost and higher utility. Probabilistic sensitivity analysis indicated that no follow-up was the better strategy in all 10,000 iterations. This strategy remained superior regardless of the NPV of the initial CT result and continued to be the better option when the incidence of missed unstable injury resulting in permanent neurologic deficits was less than 64.2% and the incidence of patients immobilized with a hard collar who still sustained cord injury was greater than 19.7%. Multiple three-way sensitivity analyses were performed. The authors concluded that MRI is not cost-effective for further evaluation of unstable injury in neurologically intact patients with blunt trauma after a negative cervical spine CT result.

Dynamic Supine Magnetic Resonance Imaging

Xu and colleagues (2017) analyzed the current evidence regarding the role of dynamic supine MRI (dsMRI) in the evaluation of cervical spondylotic myelopathy. A total of 13 studies were identified through a comprehensive literature search performed in the PubMed, Embase, and ISI databases as fulfilling the inclusion criteria and were reviewed for subject characteristics, radiographic parameters, and salient findings. Studies reviewed suggested that dsMRI was able to detect new appearance or increased grade of medullary compression in greater than or equal to 20% of patients and to demonstrate an average narrowing of the cervical canal by 20% (in comparison with the neutral position). Several additional parameters were investigated, but their clinical significance remained unconfirmed; 2 studies examined how surgical decision-making could be affected by the additional findings of dsMRI. The authors concluded that dsMRI represents an available modification of conventional static MRI and is potentially able to demonstrate pathologies that might be previously missed. They stated that evidence suggested that dsMRI can elucidate spinal cord compression with

higher sensitivity, resulting in improved diagnostic accuracy of cervical spondylotic myelopathy, which may impact surgical planning for these patients; however, more high-quality studies are needed to further establish its indications to avoid over-diagnosis with this powerful imaging technique.

Routine MRI after a Normal CT of the Cervical Spine in Obtunded or Comatose Individuals

Khanna and associates (2012) stated that the value of MRI in the evaluation of the obtunded or comatose patient with a potential neck injury is a controversial subject. Some authors have suggested that MRI of the cervical spine adds no value in the evaluation of patients with a normal CT of the neck. However, others have suggested that MRI is the gold standard for clearing the cervical spine in a clinically suspicious or unevaluable blunt trauma patient. These researchers examined their data in regard to these conflicting hypotheses. Five consecutive years of data from 17,000 patients seen at the authors' Level I trauma center yielded 512 individuals who underwent both CT and MRI of the cervical spine. Of the latter group, 150 individuals met 3 strict inclusion criteria for this study: (i) obtundation (Glasgow Coma Scale less than or equal to 13, with 94 of this group comatose [Glasgow Coma Scale less than or equal to 8]); (ii) no obvious neurologic deficits; and (iii) a normal cervical CT. The effect of MRI on the clinical management of these patients was evaluated. Among the 150 obtunded or comatose patients with a negative CT, the majority (51%) had a normal MRI. Among the patients with a positive MRI, the most common MRI-positive findings were ligamentous and soft tissue injury (81%). However, no MRI findings were deemed unstable, and no surgical intervention or change in the clinical management aside from collar immobilization of these individuals occurred after MRI. The authors concluded that the addition of a cervical MRI to the evaluation protocol of obtunded or comatose patients with an otherwise normal neurologic examination and a normal cervical CT did not provide any additional useful information to change the management of these patients.

Raza and co-workers (2013) stated that a true gold standard to rule out a significant cervical spine injury in a subset of blunt trauma patients with altered sensorium is still to be agreed upon. These investigators examined whether a clinically significant injury to the cervical spine could be ruled out in obtunded adult patients with blunt trauma based on a normal multi-detector cervical spine CT. A comprehensive database search was conducted to include all prospective and retrospective studies on blunt trauma patients with altered sensorium undergoing cervical spine multi-detector CT scans as the core imaging modality to "clear" the cervical spine. The studies used two main gold standards: MRI of the cervical spine and/or prolonged clinical follow-up. Data were extracted to report true positives, true negatives, false positives, and false negatives. A meta-analysis of sensitivity, specificity, and negative and positive predictive values (NPV and PPV) was performed using Meta Analyst Beta 3.13 software. These researchers also conducted a retrospective investigation comparing robust clinical follow-up and/or cervical spine MRI findings in 53 obtunded blunt trauma patients who previously had undergone a normal multi-detector CT scan of the cervical spine reported by a radiologist. A total of 10 studies involving 1,850 obtunded blunt trauma patients with initial cervical spine CT scans reported as normal were included in the final meta-analysis. The cumulative NPV and specificity of cervical spine CT from the 10 studies was 99.7% (95% confidence interval [CI]: 99.4 to 99.9%). The PPV and sensitivity were 93.7% (95% CI: 84.0 to 97.7%). In the retrospective review of obtunded blunt trauma patients, none were later diagnosed with a significant cervical spine injury that required a change in clinical management. The authors concluded that in a blunt trauma patient with altered sensorium, a normal cervical spine CT scan was conclusive to safely rule out a clinically significant cervical spine injury. They stated that the findings of this meta-analysis strongly supported the removal of cervical precautions in obtunded blunt trauma patients after a normal cervical spine CT; any further imaging, such as MRI of the cervical spine, should be performed on a case-by-case basis.

Smith (2014) addressed the question, "Can CT alone provide adequate clinical information to clear the cervical spine in the obtunded patient?" The author performed a search of the literature for studies that compared CT with other radiologic modalities utilized to clear the cervical spine in obtunded patients. PubMed, TRIP database, SUMSearch, Cochrane Library, and Google Scholar were the databases used. Additional sources included bibliographies of selected articles. Studies that integrated CT scans with at least one other diagnostic examination were included. A review of 11 studies and one meta-analysis encompassing 2,458 and 14,327 patients, respectively, met the inclusion criteria. The meta-

analysis generated an NPV for CT scans of 100% without evidence of acute injury, with an overall sensitivity and specificity of 99.9% each. The author concluded that these findings suggested that CT alone is a reliable clinical indicator to clear the cervical spine in obtunded patients.

Patel and colleagues (2015) noted that using the framework advocated by the Grading of Recommendations Assessment, Development and Evaluation (GRADE) Working Group, they performed a systematic review to develop evidence-based recommendations that may be used to answer the following PICO (Population, Intervention, Comparator, Outcomes) question: In the obtunded adult blunt trauma patient, should cervical collar removal be performed after a negative high-quality cervical spine (C-spine) CT result alone or after a negative high-quality C-spine CT result combined with adjunct imaging, to reduce peri-clearance events such as new neurologic change, unstable C-spine injury, stable C-spine injury, need for post-clearance imaging, false-negative CT imaging results on re-review, pressure ulcers, and time to cervical collar clearance? The protocol was registered with the PROSPERO international prospective register of systematic reviews on August 23, 2013. Eligibility criteria consisted of adult blunt trauma patients aged 16 years or older who underwent C-spine CT with axial thickness of less than 3 mm and who were obtunded using any definition. Quantitative synthesis via meta-analysis was not possible because of pre-post, partial-cohort, and quasi-experimental study design limitations and the consequential incomplete diagnostic accuracy data. Of the five articles with a total follow-up of 1,017 included subjects, none reported new neurologic changes (paraplegia or quadriplegia) after cervical collar removal. There was a worst-case 9% (161 of 1,718 subjects in 11 studies) cumulative literature incidence of stable injuries and a 91% NPV of no injury after coupling a negative high-quality C-spine CT result with 1.5-T MRI, upright X-rays, flexion-extension CT, and/or clinical follow-up. Similarly, there was a best-case 0% (0 of 1,718 subjects in 11 studies) cumulative literature incidence of unstable injuries after a negative initial imaging result with a high-quality C-spine CT. The authors concluded that in obtunded adult blunt trauma patients, they conditionally recommended cervical collar removal after a negative high-quality C-spine CT scan result alone.

Plackett and colleagues (2016) noted that the role of cervical spine MRI in the evaluation of clinically unevaluable blunt trauma patients has been called into question by several recent studies. These investigators performed a PubMed search for all studies comparing CT and MRI in the assessment of the cervical spine in patients who cannot be evaluated clinically. The radiologic findings and clinical outcomes from each study were collated for analysis. Data for 1,714 patients were available. All patients had a negative CT scan and then underwent an MRI. There were 271 (15.8%) patients who had a previously undocumented finding on MRI, with the majority (98.2%) being a ligamentous injury. Only five injuries (1.8%) resulted in surgical intervention. The authors concluded that MRI identified additional injuries; however, the vast majority were of minor clinical significance. They stated that routine MRI after a negative CT of the cervical spine is not supported by the current literature.

Endler and colleagues (2021) noted that cervical spine CT is regularly performed to exclude cervical spine injury during the initial evaluation of trauma patients. Patients with asymmetry of the lateral atlanto-dental interval (LADI) often undergo subsequent MRI to rule out ligamentous injuries. The clinical relevance of an asymmetric LADI and the benefit of additional MRI remain unclear. In a retrospective study, these researchers examined the diagnostic benefit of additional MRI in patients with blunt trauma who have asymmetry of the LADI and no other cervical injuries. Patients who underwent cervical spine CT during the initial trauma evaluation between March 2017 and August 2019 were examined. Those who underwent subsequent MRI because of LADI asymmetry of 1 mm or greater with no other signs of cervical injury were identified and re-evaluated by two readers blinded to clinical data and initial study reports regarding possible ligamentous injuries. Among 1,553 patients, 146 (9%) had LADI asymmetry of 1 mm or greater. Of these, 46 patients (mean age $\pm$ standard deviation, 39 years $\pm$ 22; 28 men; median LADI asymmetry, 2.4 mm [inter-quartile range (IQR), 1.8 to 3.1 mm]) underwent supplementary MRI with no other signs of cervical injury at initial CT; 10 of the 46 patients (22%) showed cervical tenderness at clinical examination, and 36 patients (78%) were asymptomatic. In two of the 46 patients (4%), MRI revealed alar ligament injury; both of these patients showed LADI asymmetry greater than 3 mm, along with cervical tenderness at clinical examination,

and underwent treatment for ligamentous injury. In 13 of the 46 patients (28%), signal intensity alterations of alar ligaments without signs of rupture were observed; four of these 13 patients (31%) were subsequently treated for ligamentous injury despite being asymptomatic. The authors concluded that subsequent MRI following CT of the cervical spine in trauma patients with LAD asymmetry may have diagnostic benefit only in symptomatic patients. In asymptomatic patients without proven cervical injuries, subsequent MRI showed no diagnostic benefit and may even lead to overtreatment.

Dual-Energy CT for the Evaluation of Bone Marrow Edema and Fracture Lines in Acute Vertebral Fractures

Backer and associates (2021) stated that in vertebral injuries, prompt diagnosis is essential to avoid any delays in treatment. Conventional radiography may only reveal indirect signs of fractures, such as when it is displaced; thus, to detect the presence of bone marrow edema (BME) or disc edema, adjunctive tools are needed, such as MRI or dual-energy computed tomography (DE-CT). In a systematic review and meta-analysis, these researchers examined the sensitivity, specificity, and accuracy of DE-CT of bone marrow edema and disc edema in spine injuries. Search terms included ((DECT) OR (DE-CT) OR (dual-energy CT) OR "Dual energy CT" OR (dual-energy computed tomography) OR (dual energy computed tomography)) AND ((spine) OR (vertebral)), and the PubMed, Embase, and Medline databases and the Cochrane Library and Google were used. These investigators found 1,233 articles on their preliminary search, but only 13 articles met all criteria. Data were extracted to calculate the pooled sensitivity, specificity, and diagnostic odds ratio (DOR) for analysis using R software. Within the 13 studies, 515 patients, 3,335 vertebrae, and 926 acute fractures (27.8%) defined by MRI were included. The largest cohort included 76 patients with 774 vertebrae. In 12 publications, MRI was reported for comparison. For DE-CT, the overall sensitivity was 86.2% with a specificity of 91.2% and accuracy of 89.3%. Furthermore, 5 studies reported the accuracy of CT with an overall sensitivity of 81.3%, specificity of 80.7%, and accuracy with 80.9%. Significant differences were found for specificity ($p < 0.001$) and accuracy ($p = 0.023$); however, significant inter-observer differences were reported. The authors concluded that DE-CT appeared to be a promising diagnostic tool for the detection of BME and disc edema, which can potentially replace the current gold standard, the MRI. Level of Evidence = II.

The authors stated that this review had several drawbacks. First, 4 studies were carried out by 1 research group; and individual findings for the different observers were not consistently reported. Second, only a few studies reported both the intra-observer and inter-observer reliability. Third, different kinds of MRI were used as gold standards for comparison to detect BME and disc edema. Fourth, although all studies wanted to report the accuracy of DE-CT, the studies did not use the same reconstruction algorithms or cut-offs across the board, making comparison difficult. Fifth, the largest cohort included only 76 patients.

Cavallaro and colleagues (2022) compared DE-CT and MRI for evaluating the presence and extent of traumatic BME and fracture line depiction in acute vertebral fractures. A total of 88 consecutive patients who underwent dual-source DE-CT and 3-T MRI of the spine were retrospectively analyzed. A total of 5 radiologists examined all vertebrae for presence and extent of BME and for identification of acute fracture lines on MRI and, after 12 weeks, on DE-CT series. Furthermore, image quality, image noise, and diagnostic confidence for overall diagnosis of acute vertebral fracture were assessed. Quantitative analysis of CT numbers was carried out by a 6th radiologist. Two radiologists analyzed MRI and grayscale DE-CT series to define the reference standard. For evaluating BME presence and extent, DE-CT showed high sensitivity (89% and 84%, respectively) and specificity (98% in both), and similarly high diagnostic confidence compared to MRI (2.30 versus 2.32; range of 0 to 3) for the detection of BME ($p = 0.72$). For evaluating acute fracture lines, MRI achieved high specificity (95%), moderate sensitivity (76%), and a significantly lower diagnostic confidence compared to DE-CT (2.42 versus 2.62, range of 0 to 3) ($p < 0.001$). A cut-off value of -0.43 HU provided a sensitivity of 89% and a specificity of 90% for diagnosing BME, with an overall AUC of 0.96. The authors concluded that the findings of this study demonstrated that both DE-CT and MRI provided high diagnostic confidence and image quality for the evaluation of acute vertebral fractures in general. DE-CT provided, by means of virtual non-calcium (VNCa) reconstructions, high diagnostic accuracy for evaluating the presence and

extent of vertebral BME. MRI showed moderate sensitivity and lower confidence for the depiction of fracture lines. Thus, DE-CT may represent a potential imaging alternative to MRI in specific settings of acute spinal trauma, especially in elderly or unstable patients and in cases of subtle or complex oriented fracture lines.

The authors stated that this study had several drawbacks. First, despite the relatively short examination interval, different findings between the 2 modalities concerning BME extent could not be completely excluded. BME, albeit slowly, changes over time, also depending on the adopted therapy. Second, these researchers did not separate thoracic and lumbar vertebrae, nor did they distinguish osteoporotic from non-osteoporotic patients. Age and type of vertebra have been considered as potential factors influencing bone marrow composition, and differences in bone mineral density (BMD) could influence attenuation values on VNCa images. All these factors could potentially influence BME evaluation and CT numbers on VNCa series. These researchers stated that future studies in specific population subgroups could aid in assessing the real impact of these factors in clinical routine. Third, bone marrow changes could be caused by pathologies other than trauma such as malignancy or infection. However, to the best of the authors' knowledge, other potential causes of bone marrow alteration were not present in this cohort. Finally, results and conclusions were only applicable to the vendor-specific dual-source DE-CT technique and post-processing software, as well as to the employed imaging protocols, which are routinely used in the authors' department in case of spinal trauma. Particularly in the evaluation of fracture lines, alternative MRI protocols could have performed differently, for instance by using diverse slice thickness or sequences such as T1 spoiled gradient-echo or ultra-short echo time sequences, which have recently been shown to provide image quality comparable to CT for the evaluation of certain bone pathologies of the spine.

References

The above policy is based on the following references:

1. Ahmad A, Crawford CH, 3rd, Glassman SD, et al. Correlation between bone density measurements on CT or MRI versus DEXA scan: A systematic review. *N Am Spine Soc J*. 2023;14:100204.
2. Albeck M, Hilden J, Kjaer L, et al. A controlled comparison of myelography, computed tomography and magnetic resonance imaging in clinically suspected lumbar disc herniation. *Spine*. 1995;20(4):443-448.
3. American Academy of Family Physicians. Five things physicians and patients should question. Choosing Wisely. Philadelphia, PA: American Board of Internal Medicine; 2012.
4. American Academy of Neurology. Practice Parameters: Magnetic resonance imaging in the evaluation of low back syndrome. Report of the Quality Standards Subcommittee of the American Academy of Neurology. *Neurology*. 1994;44(4):767-770.
5. American College of Physicians. Five things physicians and patients should question. Choosing Wisely. Philadelphia, PA: American Board of Internal Medicine; 2012.
6. American College of Radiology (ACR); American Society of Neuroradiology; Society of Computed Body Tomography and Magnetic Resonance. ACR-ASNR-SCBT-MR Practice Guideline for the Performance of Magnetic Resonance Imaging (MRI) of the Adult Spine. Revised 2012. Available at: https://www.asnr.org/wp-content/uploads/2017/01/3_MRI_Adult_Spine.pdf. Accessed February 6, 2026.
7. American College of Radiology (ACR), Expert Panel on Musculoskeletal Imaging. Suspected cervical spine trauma. Reston, VA: ACR; 2002.
8. Backer HC, Wu CH, Perka C, Panics G. Dual-energy computed tomography in spine fractures: A systematic review and meta-analysis. *Int J Spine Surg*. 2021;15(3):525-535.
9. Berry DB, Hernandez A, Onodera K, et al. Lumbar spine angles and intervertebral disc characteristics with end-range positions in three planes of motion in healthy people using

- upright MRI. *J Biomech.* 2019;89:95-104.
10. Bigos S, Bower O, Braen G, et al. Acute low back problems in adults. Clinical Practice Guideline No. 14. AHCPR Publication No. 95-0642. Rockville, MD: Agency for Healthcare Policy and Research (AHCPR); December 1994.
 11. Blackmore CC, Mann FA, Wilson AJ. Helical CT in the primary trauma evaluation of the cervical spine: An evidence-based approach. *Skeletal Radiol.* 2000;29(11):632-639.
 12. Boden SD. The use of radiographic imaging studies in the evaluation of patients who have degenerative disorders of the lumbar spine. *J Bone Joint Surg.* 1996;78-A(1):114-124.
 13. Butt WP. Use of gadolinium enhancement MRI in postoperative lumbar spine assessment. *Clin Radiol.* 1997;52(12):964.
 14. Callaghan B, McCammon R, Kerber K, et al. Tests and expenditures in the initial evaluation of peripheral neuropathy. 2012;172(2):127-132.
 15. Cavallaro M, D'Angelo T, Albrecht MH, et al. Comprehensive comparison of dual-energy computed tomography and magnetic resonance imaging for the assessment of bone marrow edema and fracture lines in acute vertebral fractures. *Eur Radiol.* 2022;32(1):561-571.
 16. Chen L, Wu X-Y, Jin Q, et al. The correlation between osteoporotic vertebrae fracture risk and bone mineral density measured by quantitative computed tomography and dual energy X-ray absorptiometry: A systematic review and meta-analysis. *Eur Spine J.* 2023;32(11):3875-3884.
 17. Chin CT. Spine imaging. *Semin Neurol.* 2002;22(2):205-220.
 18. Cho R, Fu R, Carrino J, et al. Imaging strategies for low-back pain: Systematic review and meta-analysis. *Lancet.* 2009;373:463-472.
 19. Daffner RH, Weissman BN, Wippold FJ II, et al; Expert Panels on Musculoskeletal and Neurologic Imaging. ACR Appropriateness Criteria suspected spine trauma [online publication]. Reston, VA: American College of Radiology (ACR); 2012.
 20. Dahabreh IJ, Hadar N, Chung M. Emerging magnetic resonance imaging technologies for musculoskeletal imaging under loading stress: Scope of the literature. *Ann Intern Med.* 2011;155(9):616-624.
 21. Davidar AD, Judy BF, Hersh AM, et al. Robot-assisted screw fixation in a cadaver utilizing magnetic resonance imaging-based synthetic computed tomography: Toward radiation-free spine surgery. Illustrative case. *J Neurosurg Case Lessons.* 2023;6(2):CASE23120.
 22. D'Andrea G, Trillo G, Roperto R, et al. Intradural lumbar disc herniations: The role of MRI in preoperative diagnosis and review of the literature. *Neurosurg Rev.* 2004;27(2):75-80; discussion 81-82.
 23. de Graaf I, Prak A, Bierma-Zeinstra S, et al. Diagnosis of lumbar spinal stenosis: A systematic review of the accuracy of diagnostic tests. *Spine.* 2006;31(10):1168-1176.
 24. Demondion X, Herbinet P, Van Sint Jan S, et al. Imaging assessment of thoracic outlet syndrome. *Radiographics.* 2006;26(6):1735-1750.
 25. Diaz JJ Jr, Cullinane DC, Altman DT, et al; EAST Practice Management Guideline Committee. Practice management guidelines for the screening of thoracolumbar spine fracture. *J Trauma.* 2007;63(3):709-718.
 26. Doyle R. Milliman Robertson Healthcare Guidelines. Vol. 4. Ambulatory Care Guidelines. Seattle, WA: Milliman; 1995; 2.54, 2.59-2.60.
 27. el Barzouhi A, Vleggeert-Lankamp CL, Lycklama a Nijeholt GJ, et al; Leiden-The Hague Spine Intervention Prognostic Study Group. Magnetic resonance imaging in follow-up assessment of sciatica. *N Engl J Med.* 2013;368(11):999-1007.
 28. Ellenberger C. MR imaging of the low back syndrome. *Neurology.* 1994;44:594-600.
 29. Endler CH, Ginzburg D, Isaak A, et al. Diagnostic benefit of MRI for exclusion of ligamentous injury in patients with lateral atlantodental interval asymmetry at initial trauma CT. *Radiology.* 2021;300(3):633-640.
 30. Eriksson S, Waldenberg C, Toren L, et al. Texture analysis of magnetic resonance images enables phenotyping of potentially painful annular fissures. *Spine (Phila Pa 1976).* 2022;47(5):430-437.

31. Expert Panel on Neurological Imaging, Hassankhani A, Freeman CW, et al. ACR Appropriateness Criteria® Acute Spinal Trauma: 2024 Update. *J Am Coll Radiol*. 2025;22(5S):S48-S66.
32. Farrell SF, Smith AD, Hancock MJ, et al. Cervical spine findings on MRI in people with neck pain compared with pain-free controls: A systematic review and meta-analysis. *J Magn Reson Imaging*. 2019;49(6):1638-1654.
33. Farris CW, Baghdanian A, Takahashi C, et al. Implementation of institutional triaging algorithms decreases head and neck MDCT use in blunt trauma. *Radiology*. 2021;298(3):622-629.
34. Fitzgerald JJ, Roberts CC, Daffner RH, et al; Expert Panel on Musculoskeletal Imaging. ACR Appropriateness Criteria® follow-up of malignant or aggressive musculoskeletal tumors [online publication]. Reston, VA: American College of Radiology (ACR); 2011.
35. Fukuda K, Kawakami G. Proper use of MR imaging for evaluation of low back pain (radiologist's view). *Semin Musculoskelet Radiol*. 2001;5(2):133-136.
36. Gerardy F, Dewandre Q, Lombard A, et al. Epidural lipomatosis: Management proposal. *Rev Med Liege*. 2021;76(2):71-76.
37. Gilbert FJ, Grant AM, Gillan MG, et al. Does early imaging influence management and improve outcome in patients with low back pain? A pragmatic randomised controlled trial. *Health Technol Assess*. 2004;8(17):iii, 1-131.
38. Gilbert FJ, Grant AM, Gillan MGC, et al. Does early magnetic resonance imaging influence management or improve outcome in patients referred to secondary care with low back pain? A pragmatic randomised controlled trial. *Health Technol Assess*. 2004;8(17):1-144.
39. Gundry CR, Fritts HM. Magnetic resonance imaging of the musculoskeletal system. Part 8. The spine, section 1. *Clin Orthop*. 1997;338:275-287.
40. Health Quality Ontario. Positional magnetic resonance imaging for people with Ehlers-Danlos syndrome or suspected craniovertebral or cervical spine abnormalities: An evidence-based analysis. *Ont Health Technol Assess Ser*. 2015;15(13):1-24.
41. Herzog R, Guyer R, Graham-Smith A, et al. Magnetic resonance imaging, use in patient with low back or radicular pain. *Spine*. 1995;20(16):1834-1838.
42. Holmes JF, Akkinepalli R. Computed tomography versus plain radiography to screen for cervical spine injury: A meta-analysis. *J Trauma*. 2005;58(5):902-905.
43. Institute for Clinical Systems Improvement (ICSI). Adult acute and subacute low back pain. Bloomington, MN: Institute for Clinical Systems Improvement (ICSI); January 2012.
44. Institute for Clinical Systems Improvement (ICSI). Adult low back pain. ICSI Health Care Guideline. Bloomington, MN: ICSI; September 2006.
45. Isaac Z, Anderson BC. Evaluation of the patient with neck pain and cervical spine disorders. UpToDate [online serial]. Waltham, MA: UpToDate; reviewed December 2014.
46. Isaac Z, Kelly HR. Evaluation of the adult patient with neck pain. UpToDate [online serial]. Waltham, MA: UpToDate; reviewed December 2019.
47. Ishikawa Y, Shimada Y, Miyakoshi N, et al. Decompression of idiopathic lumbar epidural lipomatosis: Diagnostic magnetic resonance imaging evaluation and review of the literature. *J Neurosurg Spine*. 2006;4(1):24-30.
48. Iwasaka-Neder J, Bedoya MA, Connors J, et al. Morphometric and clinical comparison of MRI-based synthetic CT to conventional CT of the hip in children. *Pediatr Radiol*. 2024;54(5):743-757.
49. Jans LBO, Chen M, Elewaut D, et al. MRI-based synthetic CT in the detection of structural lesions in patients with suspected sacroiliitis: Comparison with MRI. *Radiology*. 2021;298(2):343-349.
50. Jensen MC, Kelly AP, Brant-Zawadzki MN. MRI of degenerative disease of the lumbar spine. *Magn Reson Q*. 1994;10(3):173-190.
51. Jeon H, Min HK. Advancements in imaging techniques for early diagnosis and management of axial spondyloarthritis. *Curr Rheumatol Rep*. 2024;27(1):7.
52. Kent D, Haynor D, Lonstreth W, et al. The clinical efficacy of magnetic resonance imaging in neuroimaging. *Ann Intern Med*. 1994;120(10):856-871.

53. Kern M, Setzer M, Weise L, et al. Upright MRI after decompression of spinal stenosis and concurrent spondylolisthesis. *Neurosurg Focus*. 2019;46(5):E14.
54. Khanna P, Chau C, Dublin A, et al. The value of cervical magnetic resonance imaging in the evaluation of the obtunded or comatose patient with cervical trauma, no other abnormal neurological findings, and a normal cervical computed tomography. *J Trauma Acute Care Surg*. 2012;72(3):699-702.
55. Krabbe S, Moller JM, Hadsbjerg AEF, et al. Detection of structural lesions of the sacroiliac joints in patients with spondyloarthritis: A comparison of T1-weighted 3D spoiled gradient echo MRI and MRI-based synthetic CT versus T1-weighted turbo spin echo MRI. *Skeletal Radiol*. 2024;53(11):2459-2468.
56. Krakenes J, Kaale BR. Magnetic resonance imaging assessment of craniovertebral ligaments and membranes after whiplash trauma. *Spine*. 2006;31(24):2820-2826.
57. Lord EL, Alobaidan R, Takahashi S, et al. Kinetic magnetic resonance imaging of the cervical spine: A review of the literature. *Global Spine J*. 2014;4(2):121-128.
58. Ma H-Y, Zhang R-J, Zhou L-P, et al. Comparative effectiveness of four techniques for identifying vertebral fragility fractures among elderly patients. *Eur Radiol*. 2025;35(6):3673-3685.
59. Majumdar S. Magnetic resonance imaging and spectroscopy of the intervertebral disc. *NMR Biomed*. 2006;19(7):894-903.
60. Malhotra A, Wu X, Kalra VB, et al. Utility of MRI for cervical spine clearance after blunt traumatic injury: A meta-analysis. *Eur Radiol*. 2017;27(3):1148-1160.
61. Meinig H, Matschke S, Ruf M, et al. Arbeitsgruppe Wirbelsäulentrauma im Kindesalter der Sektion Wirbelsäule der Deutschen Gesellschaft für Orthopädie und Unfallchirurgie. Diagnostics and treatment of cervical spine trauma in pediatric patients: Recommendations from the Pediatric Spinal Trauma Group. *Unfallchirurg*. 2020;123(4):252-268.
 - Michael G, Etigunta SK, Liu AM, et al. The utilization of MRI-generated synthetic CT in the treatment of chronic bilateral spondylolysis: A case report. *JBJS Case Connect*. 2025;15(1).
63. Mintz DN. Magnetic resonance imaging of sports injuries to the cervical spine. *Semin Musculoskelet Radiol*. 2004;8(1):99-110.
64. Morbee L, Chen M, Herregods N, et al. MRI-based synthetic CT of the lumbar spine: Geometric measurements for surgery planning in comparison with CT. *Eur J Radiol*. 2021;144:109999.
65. Morbee L, Chen M, Van Den Berghe T, et al. MRI-based synthetic CT of the hip: Can it be an alternative to conventional CT in the evaluation of osseous morphology? *Eur Radiol*. 2022;32(5):3112-3120.
66. Morbee L, Vereecke E, Laloo F, et al. Common incidental findings on sacroiliac joint MRI: Added value of MRI-based synthetic CT. *Eur J Radiol*. 2023;158:110651.
67. Muchow RD, Resnick DK, Abdel MP, et al. Magnetic resonance imaging (MRI) in the clearance of the cervical spine in blunt trauma: A meta-analysis. *J Trauma*. 2008;64(1):179-189.
68. National Institutes of Health. Magnetic resonance imaging. *Natl Inst Health Consens Dev Conf Consens Statement*. 1987;6(14):1-10.
69. Nevedal AL, Lewis ET, Wu J, et al. Factors influencing primary care providers' unneeded lumbar spine MRI orders for acute, uncomplicated low-back pain: A qualitative study. *J Gen Intern Med*. 2020;35(4):1044-1051.
70. North American Spine Society. Five things physicians and patients should question. Choosing Wisely. Philadelphia, PA: American Board of Internal Medicine; 2013.
71. Owers DS, Perriman DM, Smith PN, et al. Evidence for cervical muscle morphometric changes on magnetic resonance images after whiplash: A systematic review and meta-analysis. *Injury*. 2018;49(2):165-176.
72. Papavero L, Ebert S, Marques CJ. The prevalence of redundant nerve roots in patients with lumbar spinal stenosis is body position dependent: A retrospective observational study with repeated measures design in an upright MRI scanner. *Neuroradiology* 2020;62(8):979-985.
73. Patel MB, Humble SS, Cullinane DC, et al. Cervical spine collar clearance in the obtunded adult blunt trauma patient: A systematic review and practice management guideline from

- the Eastern Association for the Surgery of Trauma. *J Trauma Acute Care Surg.* 2015;78(2):430-441.
74. Pierre-Jerome C, Arslan A, Bekkelund SI. MRI of the spine and spinal cord: Imaging techniques, normal anatomy, artifacts, and pitfalls. *J Manipulative Physiol Ther.* 2000;23(7):470-475.
 75. Plackett TP, Wright F, Baldea AJ, et al. Cervical spine clearance when unable to be cleared clinically: A pooled analysis of combined computed tomography and magnetic resonance imaging. *Am J Surg.* 2016;211(1):115-121.
 76. Quencer RM. Spine imaging. *Am J Neuroradiol.* 2000;21(1):2-8.
 77. Qureshi S, Dhall SS, Anderson PA, et al. Congress of neurological surgeons systematic review and evidence-based guidelines on the evaluation and treatment of patients with thoracolumbar spine trauma: Radiological evaluation. *Neurosurgery.* 2019;84(1):E28-E31.
 78. Raza M, Elkhodair S, Zaheer A, Yousaf S. Safe cervical spine clearance in adult obtunded blunt trauma patients on the basis of a normal multidetector CT scan -- a meta-analysis and cohort study. *Injury.* 2013;44(11):1589-1595.
 79. Resnick DK, Choudhri TF, Dailey AT, et al. Guidelines for the performance of fusion procedures for degenerative disease of the lumbar spine. Part 6: Magnetic resonance imaging and discography for patient selection for lumbar fusion. *J Neurosurg Spine.* 2005;2(6):662-669.
 80. Richmond BJ, Ghodadra T. Imaging of spinal stenosis. *Phys Med Rehabil Clin N Am.* 2003;14(1):41-45.
 81. Rothman S. The diagnosis of infections of the spine by modern imaging techniques. *Orthop Clin North Am.* 1996;27(1):15-31.
 82. Roudsari B, Jarvik JG. Lumbar spine MRI for low back pain: Indications and yield. *AJR Am J Roentgenol.* 2010;195(3):550-559.
 83. Runge VM, Muroff LR, Jinkins JR. Central nervous system: Review of clinical use of contrast media. *Top Magn Reson Imaging.* 2001;12(4):231-263.
 84. Rustagi T, Mendel E, Ferguson SA, et al. Differences in lumbar spine measures as a function of MRI posture in low back pain patients and its clinical implications. *Int J Neurosci* 2020 Sep 23:1-10..
 85. Rutkove SB. Overview of polyneuropathy. UpToDate [online serial]. Waltham, MA: UpToDate; January 2012.
 86. Rutkove SB, Tarulli A. Polyradiculopathy: Spinal stenosis, infectious, carcinomatous, and inflammatory nerve root syndromes. UpToDate [online serial]. Waltham, MA: UpToDate; reviewed January 2022.
 87. Saveyn T, Jans L, Laloo F. MRI AI use case: Synthetic CT images for fracture evaluation. *J Belg Soc Radiol.* 2022;106(1):58.
 88. Schiettecatte E, Vereecke E, Jaremko JL, et al. MRI-based synthetic CT for assessment of the bony elements of the sacroiliac joints in children. *Insights Imaging.* 2024;15(1):53.
 89. Schoenfeld AJ, Bono CM, McGuire KJ, et al. Computed tomography alone versus computed tomography and magnetic resonance imaging in the identification of occult injuries to the cervical spine: A meta-analysis. *J Trauma.* 2010;68(1):109-113; discussion 113-114.
 90. Seidenwurm D, Drayer BP, Anderson RE, et al. and the American College of Radiology. ACR appropriateness criteria for myelopathy. *Radiology.* 2000;215(Suppl):495-505.
 91. Sertic M, Parkes L, Mattiassi S, et al. The efficacy of computed tomography-guided percutaneous spine biopsies in determining a causative organism in cases of suspected infection: A systematic review. *Can Assoc Radiol J.* 2019;70(1):96-103.
 92. Shaikh N, Zhang H, Brown SHM, et al. The effect of posture on lumbar muscle morphometry from upright MRI. *Eur Spine J.* 2020;29(9):2306-2318.
 93. Shcherbakova YM, Lafranca PPG, Foppen W, et al. A multipurpose, adolescent idiopathic scoliosis-specific, short MRI protocol: A feasibility study in volunteers. *Eur J Radiol.* 2024;177:111542.
 94. Slaar A, Fockens MM, Wang J, et al. Triage tools for detecting cervical spine injury in pediatric trauma patients. *Cochrane Database Syst Rev.* 2017;12:CD011686.

95. Smith JS. A synthesis of research examining timely removal of cervical collars in the obtunded trauma patient with negative computed tomography: An evidence-based review. *J Trauma Nurs.* 2014;21(2):63-67.
96. Spinnato P, Barakat M, Lotrecchiano L, et al. MRI features and clinical significance of spinal epidural lipomatosis: All you should know. *Curr Med Imaging.* 2022;18(2):208-215.
97. Staartjes VE, Seevinck PR, Vandertop WP, et al. Magnetic resonance imaging-based synthetic computed tomography of the lumbar spine for surgical planning: A clinical proof-of-concept. *Neurosurg Focus.* 2021;50(1):E13.
98. Steffens D, Hancock MJ, Maher CG, et al. Does magnetic resonance imaging predict future low back pain? A systematic review. *Eur J Pain.* 2014;18(6):755-765.
99. Suri P, Fry AL, Gellhorn AC. Do muscle characteristics on lumbar spine magnetic resonance imaging or computed tomography predict future low back pain, physical function, or performance? A systematic review. *PM R.* 2015;7(12):1269-1281.
100. Thawait SK, Marcus MA, Morrison WB, et al. Research synthesis: What is the diagnostic performance of MRI to discriminate benign from malignant vertebral compression fractures? Systematic review and meta-analysis. *Spine (Phila Pa 1976).* 2012;37(12):E736-E744.
101. Tins BJ, Cassar-Pullicino VN. Imaging of acute cervical spine injuries: Review and outlook. *Clin Radiol.* 2004;59(10):865-880.
102. Townsend DC, Purohit N, Giannoulis K, Shtaya A. Presentation, management, and outcome of primary leiomyosarcoma of the spine: A systematic review. *World Neurosurg.* 2022;163:25-35.
103. University of Michigan Health System (UMHS). Acute low back pain. UMHS Clinical Guideline. Ann Arbor, MI: University of Michigan Health System; April 2003.
104. Upadhyay J, Iwasaka-Neder J, Golden E, et al. Synthetic CT assessment of lesions in children with rare musculoskeletal diseases. *Pediatrics.* 2023;152(2):e2022061027.
105. van der Kolk BBYM, Slotman DJ, Nijholt IM, et al. Bone visualization of the cervical spine with deep learning-based synthetic CT compared to conventional CT: A single-center noninferiority study on image quality. *Eur J Radiol.* 2022;154:110414.
106. Vanichkachorn JS, Vaccaro AR. Thoracic disk disease: Diagnosis and treatment. *J Am Acad Orthop Surg.* 2000;8(3):159-169.
107. Veiga JRS, Mitchell K. Cervical spine clearance in the adult obtunded blunt trauma patient: A systematic review. *Intensive Crit Care Nurs.* 2019;53:109-110.
108. Vereecke E, Morbée L, Laloo F, et al. Anatomical variation of the sacroiliac joints: An MRI study with synthetic CT images. *Insights Imaging.* 2023;14(1):30.
109. Walker PB, Sark C, Brennan G, et al. Spinal epidural lipomatosis: A comprehensive review. *Orthop Rev (Pavia).* 2021;13(2):25571.
110. Washington State Department of Labor and Industries, Office of the Medical Director. Criteria for MRI of the lumbar spine. Olympia, WA: Washington State Department of Labor and Industries; June 1999.
111. Washington State Department of Labor and Industries, Office of the Medical Director. Cauda equina. Olympia, WA: Washington State Department of Labor and Industries; June 1999.
112. Weber U, Zubler V, Zhao Z, et al. Does spinal MRI add incremental diagnostic value to MRI of the sacroiliac joints alone in patients with non-radiographic axial spondyloarthritis? *Ann Rheum Dis.* 2015;74(6):985-992.
113. Weiß T, Disch AC, Kreinest M, et al; Arbeitsgruppe Wirbelsäulentrauma im Kindesalter der Sektion Wirbelsäule der Deutschen Gesellschaft für Orthopädie und Unfallchirurgie. [Diagnostics and treatment of thoracic and lumbar spine trauma in pediatric patients: Recommendations from the Pediatric Spinal Trauma Group.] *Unfallchirurg.* 2020;123(4):269-279.
114. Wilkinson LS, Elson E, Saifuddin A, et al. Defining the use of gadolinium enhanced MRI in the assessment of the postoperative lumbar spine. *Clin Radiol.* 1997;52(7):530-534.
115. Willesen ST, Hadsbjerg AE, Moller JM, et al. MRI-based synthetic CT: A new method for structural damage assessment in the spine in patients with axial spondyloarthritis -- A comparison with low-dose CT and radiography. *Ann Rheum Dis.* 2024;83(6):807-815.

116. Wu X, Malhotra A, Geng B, et al. Cost-effectiveness of magnetic resonance imaging in cervical spine clearance of neurologically intact patients with blunt trauma. *Ann Emerg Med*. 2018;71(1):64-73.
117. Xu N, Wang S, Yuan H, et al. Does dynamic supine magnetic resonance imaging improve the diagnostic accuracy of cervical spondylotic myelopathy? A review of the current evidence. *World Neurosurg*. 2017;100:474-479.
118. Yamada Y, Demura S, Kato S, et al. Anatomical characteristics and variations of thoracic segmental arteries with 3-dimensional CT analysis. *Spine (Phila Pa 1976)*. 2022;47(22):E617-E624.
119. Yang Y, Ma L, Li L, Liu H. Primary leiomyosarcoma of the spine: A case report and literature review. *Medicine (Baltimore)*. 2017;96(9):e6227.



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